MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Erasmus Mundus Master in Astrophysics and Space Science

Master's thesis

Reconstructing Photometric Lightcurves of Active Galactic Nuclei using Neural Stochastic Flows

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УНИВЕРЗИТЕТ У БЕОГРАДУ
МАТЕМАТИЧКИ ФАКУЛТЕТ

Ana Cecilia Luis Ramirez

Реконструкција фотометријских светлосних кривих активних галактичких језгара применом стохастичких Neural Flow модела

Master Thesis

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1 Background

Compared to ordinary galaxies, some have extraordinarily bright central regions, which can sometimes even outshine the host-galaxy emission by many orders of magnitude; these galaxies are called *active galaxies* and their nuclei are referred to as *active galactic nuclei* (AGN). The physical processes necessary to create such high luminosities ($L = 10^{43-48}$erg s^{-1}; e.g., Lusso et al. 2012) within such a small volume were a subject of debate for years, but now are widely accepted to be connected to accretion onto the galaxies' central black holes (BHs; Salpeter 1964; Zeldovich 1964). These accreting objects have masses $M_{BH} \sim 10^{6-10} M_\odot$ (e.g., Peterson and Horne 2004) and an Eddington ratio exceeding λ_{Edd}[1] $\approx 10^{-5}$. This threshold, chosen somewhat arbitrarily, excludes galactic nuclei with no significant accretion activity, such as that of the Milky Way, while still covering their full range of observed luminosities (Netzer 2015).

The "unified model" (Antonucci 1993; Urry and Padovani 1995) discussed in Section 1.1, posits the black hole at the center of the AGN, surrounded by several distinct components, including an accretion disk, gas and dust clouds, and, in some cases, a relativistic jet (Figure 1.4). Each of these regions contributes emission into different wavelength regimes, resulting in these objects having detectable emission across the whole electromagnetic spectrum, as illustrated by the spectral energy distribution (SED) in Figure 1.1.

The gravitational energy released as the circumnuclear material falls inwards and loses angular momentum forms the accretion flow around the black hole, which acts as the primary source of AGN continuum emission. The disk gets heated up via viscous heating (Peterson 1993), producing a thermal spectrum that has a blackbody temperature peaking in the ultra violet (UV; $\approx 10 - 400$ nm) for a "typical" AGN (i.e., with $M_{BH} \approx 10^8 M_\odot$; $\lambda_{Edd} \approx 0.1$); this is often referred to as the "big blue bump" (Harrison 2014; Figure 1.1)

The thermal disk photons, also called *seed photons*, are Comptonized by highly energetic electrons that are located at the *corona* (§1.1.2) to form the X-ray power-law spectrum observed in Figure 1.1. These high energy photons can also reflect off the accretion disk and/or the *torus* (§1.1.4), producing the additional hard X-ray "reflection" component. In a majority of AGN X-ray spectra, there is an additional "soft excess" (Singh et al. 1985; Arnaud et al. 1985) at energies less than ~ 2 keV. This emission is above what would be expected from simple accretion disk models and its origin is still a topic of discussion. Two proposed models (which could be simultaneously true) to explain this soft excess are: relativistically blurred ionized reflection from the

[1]The ratio L/L_{Edd}, where L is the bolometric luminosity and $L_{Edd} = 1.5 \times 10^{38} M_{BH}/M_\odot$ erg s^{-1} is the Eddington luminosity for a solar composition gas. This is the maximum isotropic luminosity a body can achieve while maintaining balance between radiation pressure and gravitational force.

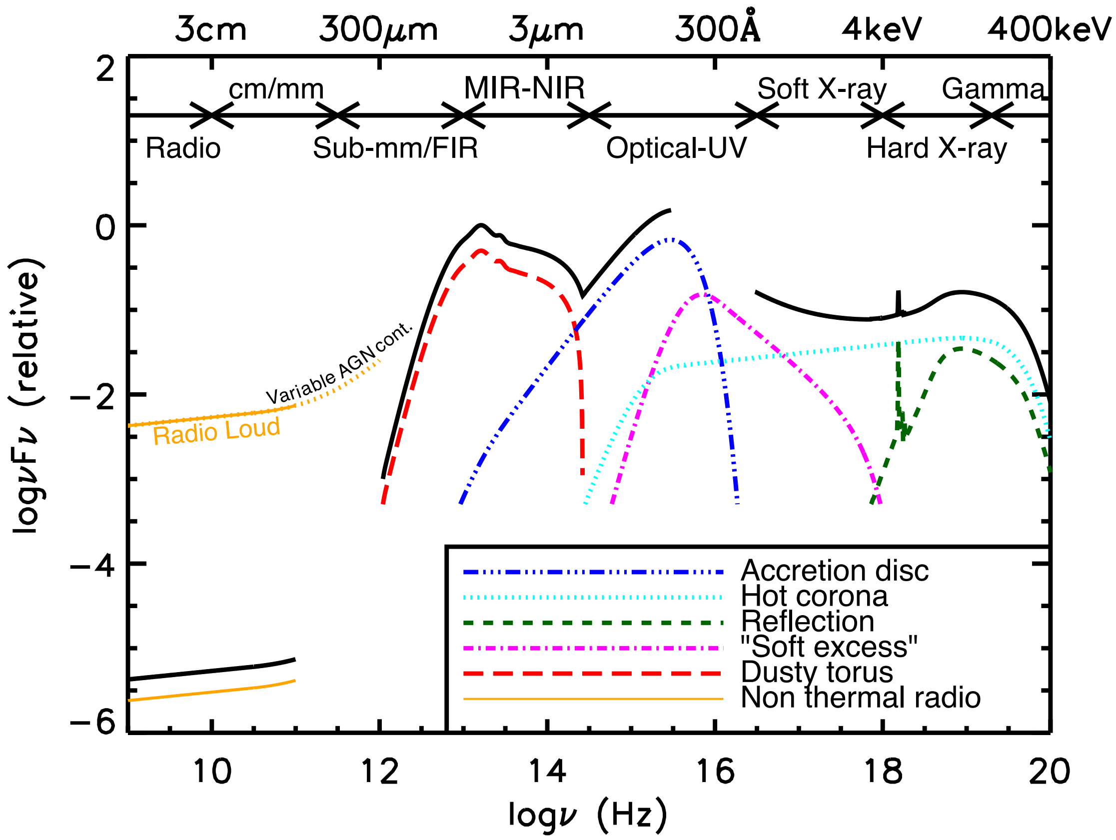


Figure 1.1: AGN SED from radio frequencies to hard X-ray, loosely based on the observed SEDs of radio-quiet quasars (e.g., Elvis et al. 1994; Richards et al. 2006). The black solid curve represents the total emission and the colored curves (shifted down for clarity) represent the individual constituents. The intrinsic shape of the SED in the mm-far infrared (FIR) regime is uncertain; however, it is credited to have a minimal contribution (to an overall galaxy SED) compared to star formation (SF), except in the most intrinsically luminous quasars and powerful jetted AGN. It should be noted that the relative contribution of each of the components can vary dramatically for different AGN types. Adapted from Harrison (2014).

inner accretion disc (e.g., Crummy et al. 2006), or Comptonization in a "warm corona" located near the surface of the accretion flow (e.g., Ballantyne and Xiang 2022).

The infrared (IR; $\approx 1 - 1000$ μm) band, is mostly sensitive to the reprocessed emission of the accretion disk by the torus. Although the shape of this IR emission varies from source to source, the peak of this dusty component is usually around $\lambda \approx 20 - 50$ μm and falls steeply at longer wavelengths (Harrison 2014). In the sub-mm/far-infrared (FIR), or sometimes even in the mid-infrared (MIR), stellar heated emission from cooler dust dominates over AGN heated emission (Lutz 2014).

AGN cover a wide range of radio luminosities, which can differ by several orders of magnitude between radio-quiet (RQ) and radio-loud (RL) sources, now more aptly referred to as non-jetted and jetted, respectively [2] (orange lines in Figure 1.1). The latter type makes up only a minority of the AGN population (e.g., Padovani 2011) with objects that present highly collimated, relativistic radio *jets* (§1.1.5) and extended radio structures such as lobes and hotspots, that together emit strong non-thermal radiation and, occasionally, a γ-ray jet. On the other hand, the spectrum of non-jetted objects is dominated by thermal emission in most bands and the origin

[2] While RL/RQ has been the historical terminology, radio-quiet sources are not truly radio-silent but rather "radio-faint" and they are, however, γ-ray quiet (Padovani 2017).

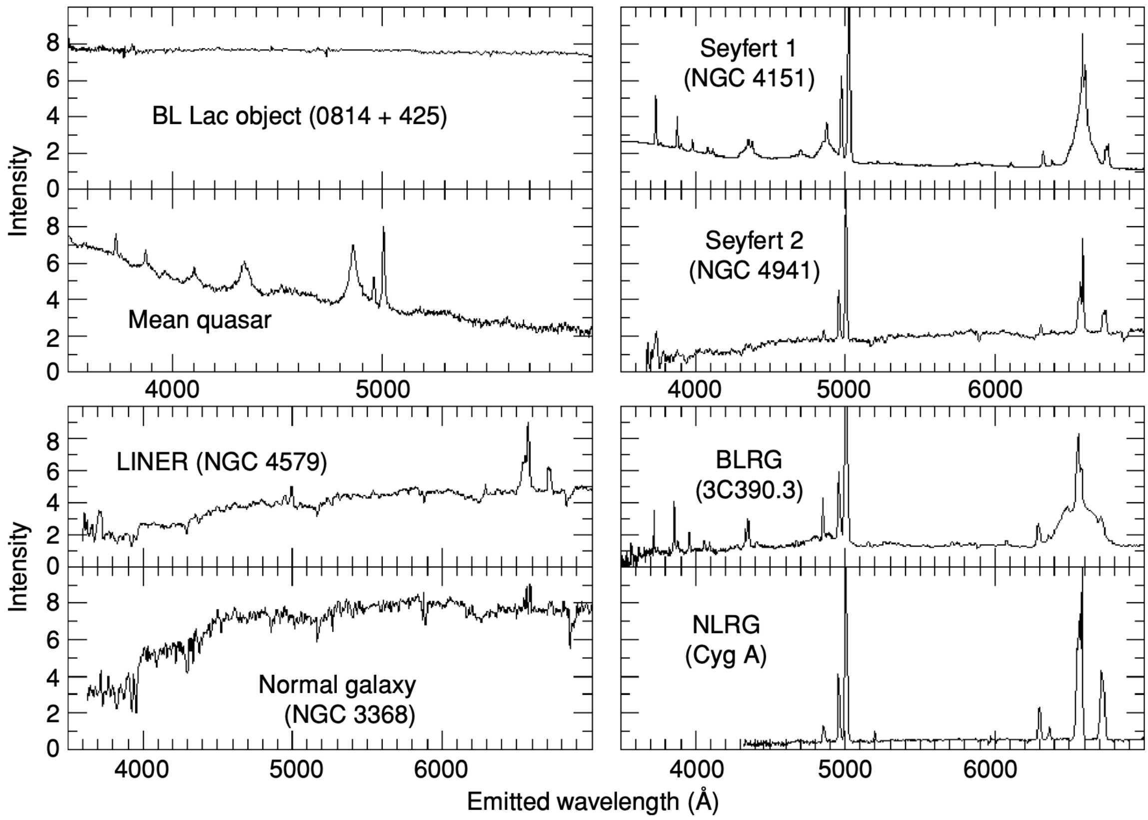


Figure 1.2: Optical spectra of representative AGN classes, from the (essentially featureless) BL Lac objects (Top left) through the radio galaxies and QSOs (Seward and Charles 2010). Note the presence of both broad and narrow emission lines in quasars, Seyfert 1 galaxies and broad-line radio galaxies (BLRG), while Seyfert 2 galaxies, low-ionisation nuclear emission region (LINER) galaxies and narrow-line radio galaxies (NLRG) only show narrow permitted and forbidden lines. A normal galaxy spectrum (Bottom left) is shown for comparison.

of their radio emission has not been clearly established. Some possibilities include coronal activity or synchrotron radiation from particles accelerated in shocks (see e.g., Laor and Behar 2008; Ishibashi and Courvoisier 2011).

1.1 Unified Model of AGN

Because different wavebands probe different physical elements, AGN selected at particular wavelengths often exhibit distinct properties, leading to a rich phenomenological classification (Figure 1.2). Reality, however, is much simpler because now we know that most of these classes differ from each other only by a small number of parameters, such as: orientation, accretion rate, the presence of jets, and possibly the host galaxy and environment (see Padovani et al. (2017) and references there-in). This suggests that the numerous brands do not necessarily correspond to fundamentally different physical objects, but instead motivate a unifying framework capable of explaining their apparent diversity.

This unification idea arose shortly after the discovery of *quasars* (QSOs[3]), when the observed

[3]The term quasar historically stands for "quasi-stellar radio source"; throughout this thesis it is used more broadly to refer to very luminous AGN.

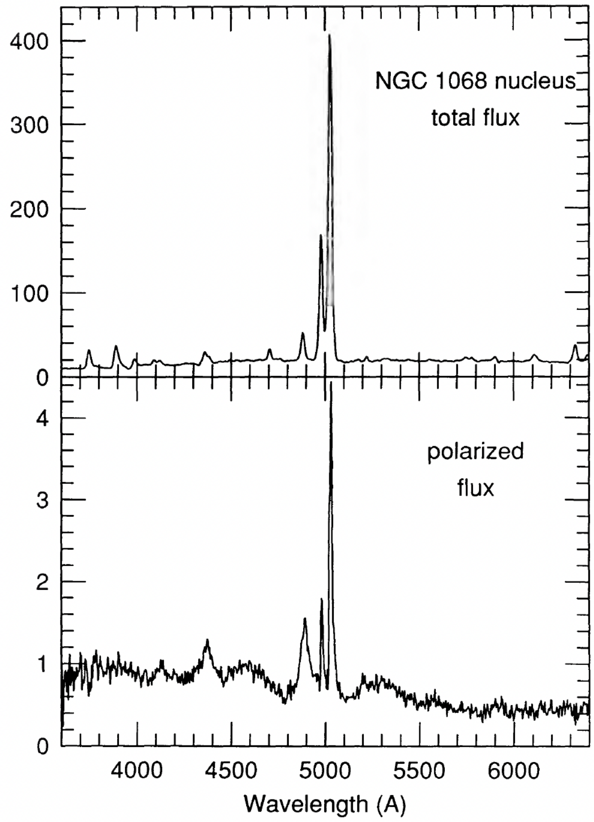


Figure 1.3: Optical spectra of the nucleus of NGC 1068 (Miller, Goodrich, and Mathews 1991). The total spectrum (Top) shows the narrow emission lines expected for Type 2 AGN, while the polarized flux spectrum (Bottom) reveals the broad permitted emission lines characteristic of unobscured Type 1 AGN.

variety of AGN prompted questions about the role of the observer's orientation relative to the AGN axis. Blandford and Rees (1978) proposed that BL Lac objects were radio galaxies viewed down the axis of a relativistic jet, where *relativistic beaming* (§1.1.5) would cause the non-thermal continuum to appear very bright and the emission lines weak in comparison. The same object, viewed from the side, would have normal emission-line equivalent widths, and the radio structure would be dominated by the extended lobes rather than the core (Shields 2004).

Further evidence came from spectropolarimetry, the measurement of the polarization state of the incoming light from a source as a function of wavelength. Seyfert galaxies (Seyfert 1943) are primarily divided into two types based on their optical spectra: Type 1 objects exhibit both broad permitted emission lines such as Hα and Hβ (FWHM $>$ 1000 km s^{-1}) and narrow forbidden lines such as [O III] and [Ne III] (FWHM $<$ 1000 km s^{-1}), while Type 2 objects show only the narrow component. Crucially, both types share the same high- and low-ionization narrow forbidden lines with similar line ratios, suggesting they are powered by the same intrinsic engine.

Antonucci and Miller (1985) confirmed this by analyzing the polarized flux of NGC 1068, a Seyfert 2 galaxy[4], resembled that of a Seyfert 1 spectrum, as can be seen in Figure 1.3. This was interpreted as the broad-line region (BLR; §1.1.3) and the central continuum source being blocked from direct view by a dusty torus, while free electrons above the nucleus scatter a fraction of the nuclear light towards the observer, polarizing it in the process. The broad lines had gone unnoticed in direct light because this scattered component was too faint relative to the

[4]It should be noted that a subset of Type 2 AGN ($\sim 30\%$), sometimes referred to as "true Type 2" objects, show no detectable broad lines even in polarized light and little X-ray absorption (see Netzer 2015 and references there-in).

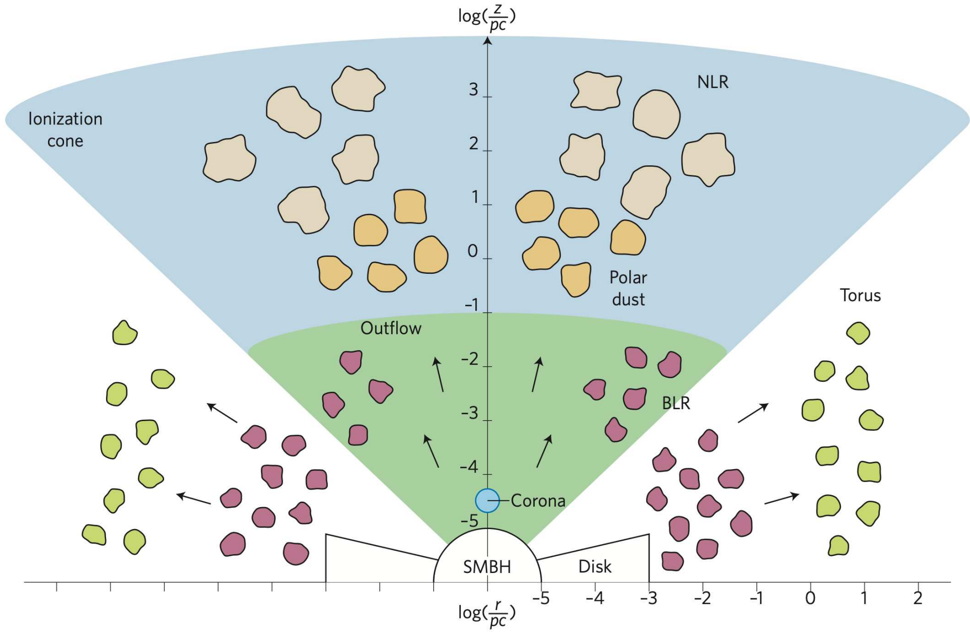


Figure 1.4: Overview of the innermost regions around a non-jetted accreating supermassive black hole (SMBH). Typical scales along the polar and equatorial directions (for nearby AGN) are shown. Different colors depict regions with different densities or compositions. Taken from Ramos Almeida and Ricci 2017.

strong narrow-line emission from the narrow-line region (NLR; §1.1.6), which extends beyond the obscuring torus and is therefore visible regardless of orientation.

Multiple efforts such as those recounted above (e.g., Antonucci 1984; Orr and Browne 1982) were fundamental to the development of the "favoured" unified model of AGN introduced in this section. Whilst it has been succesful in homogenizing many kinds of active nuclei, recent works point to the conclusion that a picture based primarily on orientation and obscuring material is incomplete (e.g., Bianchi, Maiolino, and Risaliti 2012). Nonetheless, the major aspects of the model are sufficient for the purposes of this thesis; in the sections below, the primary components are explained briefly, while additional features such as polar dust and large-scale outflows (also visible in Figure 1.4) are discussed shortly when relevant but are not treated as separate subsections.

1.1.1 Accretion Disk

The standard framework of the optically thick, geometrically thin accretion disk (Shakura and Sunyaev 1973) described above applies to SMBHs with moderate to high Eddington ratios ($10^{-2} \lessapprox \lambda_{Edd} \lessapprox 1$; Ricci 2026), where the accretion flow is considered radiatively efficient. This scheme describes the disk as a composition of Keplerian rings, nested and coupled to each other by viscous forces, likely driven by turbulence and magnetic instabilities, the magneto-rotational instability (MRI; Balbus and Hawley 1998) being the most widely quoted. The outer, slower ring brakes the inner faster one, transporting the angular momentum outwards and allowing material to spiral inward. Friction between the rings produces heat that is dissipated as blackbody emission with a characteristic maximum temperature of $T \sim 10^5$ K for the "typical" AGN described above (Netzer 2013), yielding the multi-temperature spectrum depicted in Figure 1.1. The continuum emission of the accretion disk varies with time depending on the dynamics of the infalling matter (see Chapter 2).

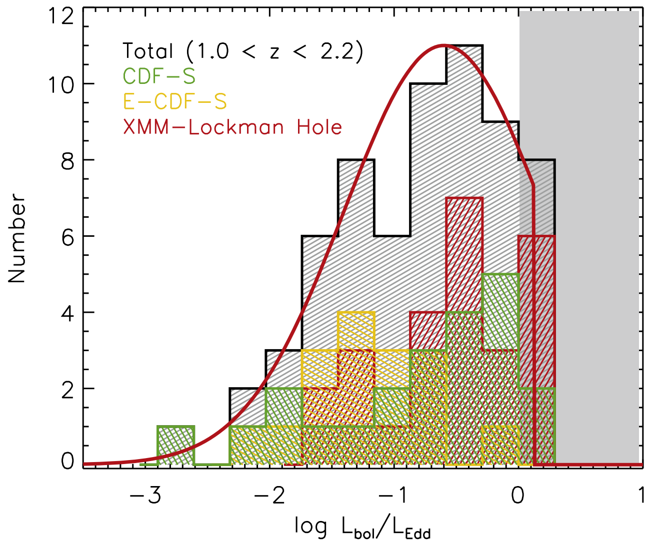


Figure 1.5: Eddington ratio distribution of broad-line AGNs at $1.0 < z < 2.2$. Different X-ray surveys are represented as different color histograms, and the black histogram pictures all the combined distributions. The gray shade illustrates the Eddington limit ($\lambda_{Edd} = 1$) and the red solid line corresponds to a log-normal fit with a peak of $\log L/L_{Edd} = -0.6$ (Suh et al. 2015).

Most QSOs in the local universe are thought to accrete through thin disks, yet contrasting accretion scenarios are possible depending on the Eddington ratio (Figure 1.5). At $\lambda_{Edd} \lesssim 10^{-2}$, the disk becomes optically thin and with an Advection Dominated Accretion Flow (ADAF, Narayan and Yi 1994), radiating weakly and with minimal optical/UV emission. Conversely, at $\lambda_{Edd} \gtrsim 1$, theoretical works have predicted that the SMBHs accrete rapidly through "slim" (or "thick") disks, showing very different SEDs than their slower counterparts and powerful outflows (e.g., Kubota and Done 2018; Giustini and Proga 2019).

1.1.2 Corona

Numerical simulations of magnetohydrodynamical accretion disks have shown that the dissipation of the magnetic energy built up by the MRI naturally produces a hot, active corona located within a few gravitational radii ($R_g = GM/c^2$) of the accretion disk (Balbus and Hawley 1998; Miller and Stone 2000; Fabian et al. 2009). This compact region of ionized gas reaches temperatures of $T \sim 10^9$ K (Kara and García 2025), and its rapid variability (see Chapter 2) is consistent with such a small emitting region. The corona contributes approximately 5% of the AGN bolometric output (Ricci 2026), a fraction that decreases with increasing λ_{Edd}, with lower λ_{Edd} systems showing stronger X-ray emission relative to the disk (Padovani et al. 2017).

1.1.3 Broad-line Region

The broad-line region consists of high density gas clouds ($\gtrsim 10^9$ cm^{-3}) predominantly in Keplerian motion at luminosity-dependent distances of $0.01 - 1$ pc from the central black hole (Netzer 2015), well within the *dust sublimation radius* (see §1.1.4). The BLR is therefore dust free, emitting broad permitted and semi-forbidden emission lines with velocity widths ranging from $1,000$ to $20,000$ km s^{-1} (Ricci 2026), wideness caused by *Doppler broadening*[5]. Ionizing photons from the

[5]Doppler broadening refers to the superposition of emission from clouds moving at different directions/velocities along the line of sight produces a single broadened line whose width reflects the velocity dispersion of the gas.

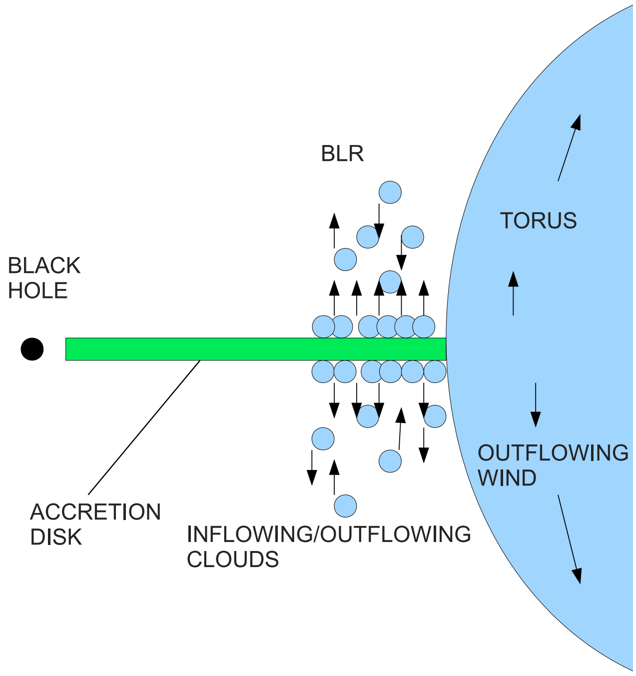


Figure 1.6: Schematic of the BLR geometry and its location relative to the accretion disk and dusty torus. The figure also illustrates the failed wind scenario of Czerny and Hryniewicz (2011): dust forms in the atmosphere of the inner accretion disk at radii where $T_{eff} < T_{sub}$, is launched as a wind, but then is suppressed by the radiation from the central source and partially falls back, giving rise to the inflowing and outflowing clouds that make up the BLR.

accretion disk and corona illuminate these clouds, which subsequently exhibit ionization stratification with respect to radius, with higher ionization species located closer to the central source (Netzer 2013). The physical origin of the BLR remains an open question, some of the models presented in the literature include radiation pressure driven disk winds (Murray et al. 1995) and failed dust-driven wind from the disk surface (Figure 1.6; Czerny and Hryniewicz 2011). Notably, the distance of the BLR from the central source scales with the AGN luminosity as $R_{\rm BLR} \propto L^{1/2}$ (Kaspi et al. 2000), a relationship that is fundamental in reverberation mapping studies (§2.1.1).

1.1.4 Dusty Torus

As posed previously, the presence of an assymetric "dusty obscurer" has successfully unified apparently different AGN classes. When observed edge-on (i.e., at high inclination angles), this torus[6] obscures the innermost regions of the AGN and hides the broad emission lines; conversely, when viewed face-on (at lower inclination angles), there is a clear line of sight to the BLR and accretion disk (Seyfert 1 vs Seyfert 2 in Figure 1.2). This structure is conceptualized as an optically thick structure surrounding the central engine at scales larger than that of the BLR and luminosity dependent dimensions ranging from $0.1 - 10$ pc (Netzer 2015). Its inner boundary coincides is the dust sublimation radius ($R_{\rm sub}$), beyond which the dust can survive the radiation from the central engine. This radius corresponds to the distance at which the dust temperature reaches the sublimation threshold ($T_{sub} \sim 1400 - 1800$ K for silicate and graphite grains respectively; e.g., Barvainis 1987; Netzer 2015).

[6]For convenience, we keep referring to this obscuring matter as the torus despite the fact that it could have a different (and more complex) geometrical distribution (e.g., Nenkova et al. 2008; Stalevski et al. 2012).

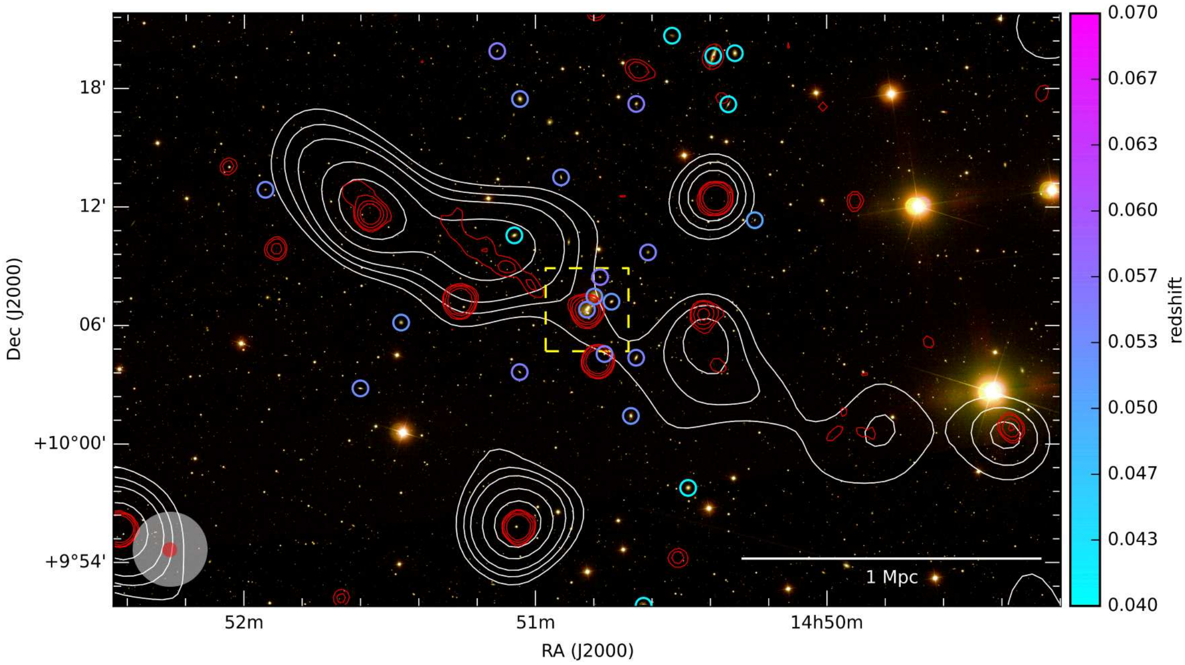


Figure 1.7: SDSS optical image (composite from bands g, r and i) of the galaxy group UGC 9555 overlaid with NVSS contours (red) and LOFAR MSSS radio contours (white) revealing the size of the giant radio galaxy (GRG). The white bar indicates a physical scale of 1 Mpc, illustrating how AGN jets can extend well beyond the boundaries of their host galaxies. Adapted from Clarke et al. 2017.

In addition to being responsible for the IR reprocessed emission on the AGN SED (Figure 1.1), the torus acts as a gas and dust reservoir that regulates the material available for accretion onto the black hole. Its geometry, believed to contain a central opening, also collimates the AGN radiation into bi-conical ionization cones along the polar direction (Pogge 1988), which illuminate the NLR (§1.1.6). Furthermore, AGN drive large scale winds and outflows along these directions, powered by radiation and magnetic pressure from the central engine, which can interact with and regulate the host galaxy interstellar medium (Ricci 2026).

1.1.5 Relativistic Jets

Jets are thought to be powered by the extraction of rotational energy from the SMBH via strong electromagnetic fields (Blandford and Znajek 1977), and can extend to Mpc scales from their host galaxy (Figure 1.7; Clarke et al. 2017).

Their launch point coincides with the location of the corona, in the immediate vicinity of the accreting object, and their apparent velocities range from mildly to highly relativistic motion. The broadband emission of jets is characterised by a double-peaked SED: the low-energy peak, spanning radio to optical wavelengths, arises from synchrotron radiation of relativistic electrons spiralling in the jet magnetic field, while the high-energy peak, from X-ray to γ-ray energies, is thought to result from inverse-Compton scattering of soft photons from the BLR and accretion disk by the same relativistic electrons (Netzer 2013).

Their observed properties depend strongly on orientation because relativistic beaming could make an approaching jet appear significantly brighter than a receding one. This effect, known as *Doppler boosting*, arises when the jet plasma has bulk relativistic motion that concentrates the emitted radiation into a narrow cone in the direction of motion, amplifying the observed flux and variability for jets viewed close to face-on and explainig the extreme characteristics observed

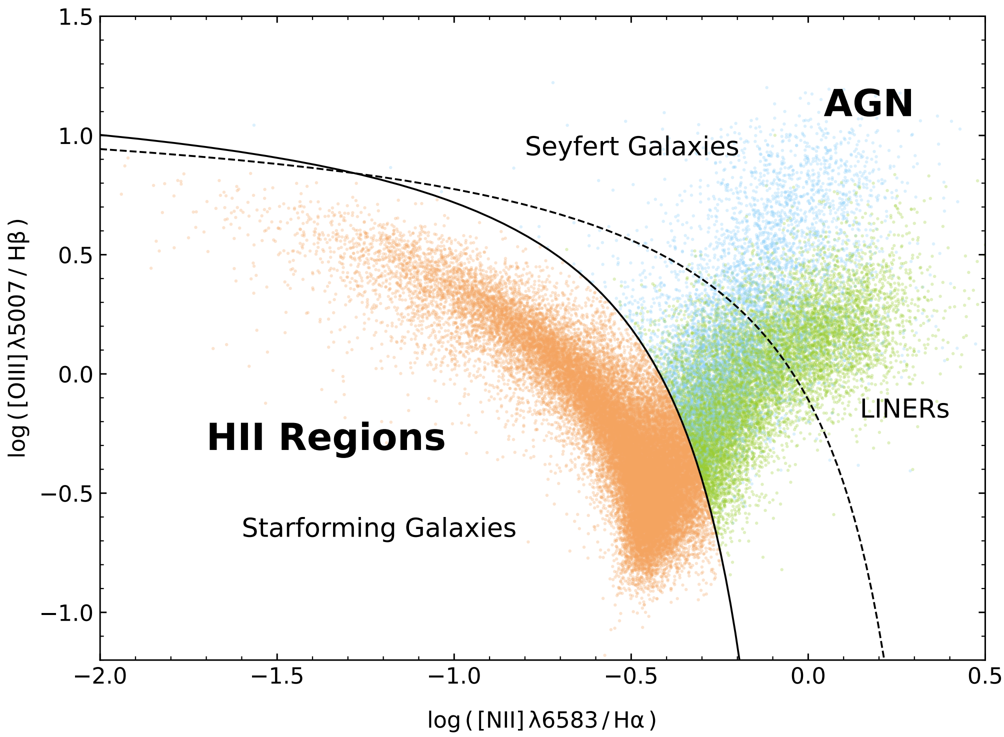


Figure 1.8: Select galaxy objects from the Sloan Digital Sky Survey (SDSS) catalog shown in the BPT diagram. It plots the intensity of the [OIII] $\lambda5007$ forbidden transition relative to that of the Hβ line versus the ratio of [NII] $\lambda6583$ to Hα on logarithmic axes. Using this to infer ionization states and abundances of the emitting medium enables us to reconstruct the origin of the dominant radiation in the galactic host, which has led to various approaches for the classification of these objects. Depicted is the theoretical delimiter by Kewley et al. (2001) as well as the empirical line by Kauffmann et al. (2003) to distinguish predominantly starforming galaxies from AGN. Additionally, the Schawinski et al. (2007) boundary sorts AGN into efficiently accreting Seyfert and inefficient LINER subtypes. Courtesy of Fritz A. Agildere.

on blazars (Urry and Padovani 1995). Jets are also thought to play a role in galaxy evolution by injecting energy into the surrounding environment, a process referred to as radio-mode feedback (Fabian 2012).

1.1.6 Narrow-line Region

This assymetric region is extended enough to be resolved optically (e.g., Fischer et al. 2013; Polack et al. 2024), spanning hundreds or even thousands parsecs along the ionization cones (Figure 1.4). The NLR, just as the BLR, is photoionized by the central engine and exhibits optical emission lines, kinematically broadened with typical widths of 10^{2-3} km s^{-1} (Bennert 2005), with most lines falling within the $350 - 500$ km s^{-1} range (Ricci 2026).

Compared to the BLR, the NLR has a lower electron density ($\sim 10^{2-3}$ cm^{-3}), allowing forbidden transitions to occur without being collisionally suppressed. The presence of these forbidden lines, together with the high ionization state of the gas, can trace AGN activity and makes the NLR emission diagnostically distinct from that of star-forming regions, a separation commonly visualised using emission line ratio diagrams such as the Baldwin-Phillips-Terlevich (BPT) diagram (Baldwin, Phillips, and Terlevich 1981) (Figure 1.8).

2 Variability of AGN

The first quasar discovered, 3C 273 (Edge et al. 1959), had been documented in photographic plates at least as early as 1887 (Slavcheva-Mihova et al. 2013). This emblematic radio-loud object showed variable emission in the optical and radio bands on timescales from days to years (Smith and Hoffleit 1963; Dent 1965), indicating that the size of the emitting region spanned only a few light-days in size. This compactness, coupled with the finding of its distant redshift (Schmidt 1963), was very difficult to explain by dynamics that operate in normal galaxies, and variability thus became one of the key arguments that the enormous luminosities of quasars had to originate from an exceptionally compact and energetic central engine (Salpeter 1964; Zeldovich 1964).

Now we know that variability in all wavelengths is an ubiquitous property of AGN, and so, its studies have been essential in understanding the physics of their nuclear regions, which, in general, cannot be resolved by current telescopes. This had been strictly true only until the observations of Sgr A* and M87 (The Event Horizon Telescope Collaboration et al. 2019) and, more recently, with the GRAVITY interferometer on the VLT resolving the outer BLR in 3C 273 (Gravity Collaboration et al. 2018); still, these surveys are limited to massive sources in the very nearby universe. For the vast majority of AGN, variability studies therefore rely on the *light curve*, a time series recording how the brightness (or flux) of the AGN, measured within a given wavelength band, varies over a monitoring period. The time scales, spectral changes, and the correlations and delays between variations extracted from these light curves provide crucial information on the nature and location of these components and on their interdependencies (Ulrich et al. 1997).

While variability-based methods are, generally speaking, more effective than other methods (e.g., SED-based) at providing a clearer signature of the nuclear activity against host galaxy emission (Yu et al. 2025), deriving the features and their uncertainties from an irregularly sampled light curve is very challenging; thereon, high quality light curves are indispensable to enable its robust characterization (Kelly et al. 2014).

2.1 Physical Origin and Timescales

Early campaigns quickly revealed that AGN vary much more rapidly in X-rays than in the optical (Winkler and White 1975), sometimes on timescales as short as a few minutes (McHardy et al. 1981), and that variability at different wavelengths is correlated (Ulrich et al. 1997). This stochastic behaviour can generally be explained by the *propagating fluctuations* theory (Lyubarskii 1997).

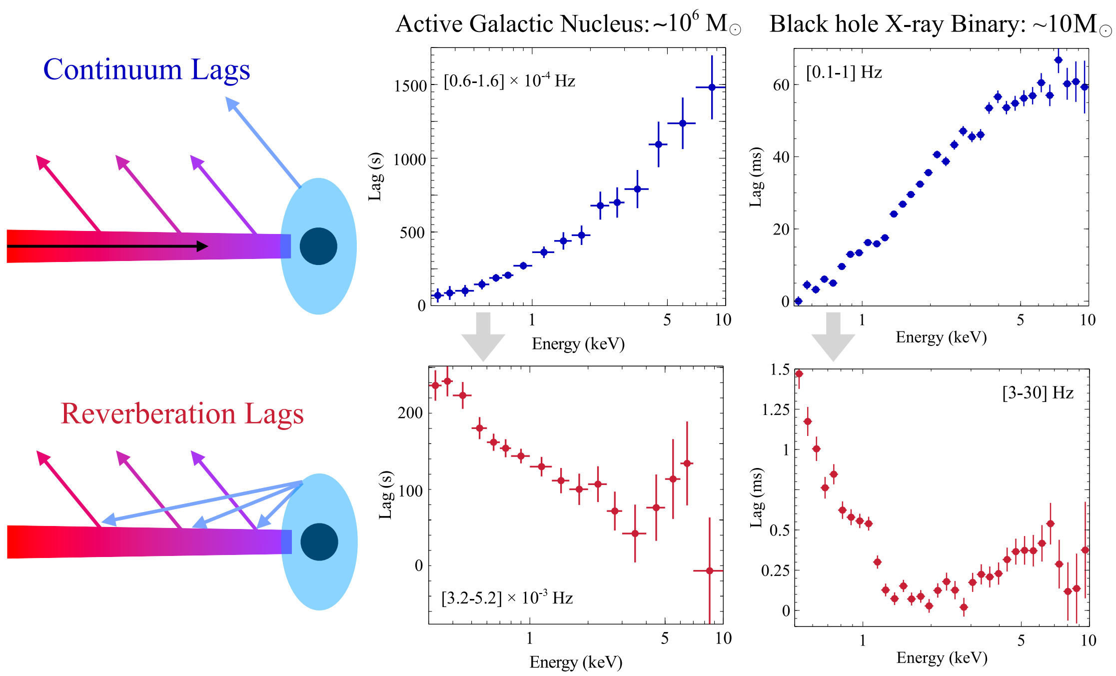


Figure 2.1: Frequency-resolved X-ray time lags as a function of energy for an AGN ($\sim 10^6\,M_\odot$, left) and a black hole X-ray binary ($\sim 10\,M_\odot$, right), illustrating the universality of the two lag regimes. At low Fourier frequencies (top, blue), hard X-rays lag soft ones, consistent with inward-propagating accretion fluctuations. At high Fourier frequencies (bottom, red), soft X-rays lag hard ones, consistent with disk reverberation. Taken from Kara and García 2025.

Local fluctuations in the disk's surface density, presumably driven by the MRI (§ 1.1.1), occur at all radii and propagate inwards at the local viscous timescale[1]. As these fluctuations travel, they modulate the local mass accretion rate at progressively smaller radii, coupling the variability of the outer disk to that of the hotter inner regions and corona, where the shorter local viscous timescale drives faster fluctuations in the emitted luminosity (Uttley et al. 2023). The X-ray variability produced in the innermost regions (i.e., in the corona) then irradiates the outer accretion disk and is reprocessed, driving the optical and UV variability observed at longer wavelengths with a characteristic time delay (Padovani et al. 2017), a *reverberation*, as discussed below.

Naturally, both propagating fluctuation lags and reverberation lags are simultaneously present in the data, but dominate at different temporal frequencies and have opposite energy dependences (Figure 2.1): at low frequencies ($\nu \lesssim 10^{-5}$ Hz for a $10^6\,M_\odot$ black hole, or equivalently, $\gtrsim 1$ day), hard X-ray variations lag soft ones as accretion fluctuations propagate inward; at high frequencies ($\nu \gtrsim 10^{-3}$ Hz, timescales of minutes), soft X-rays lag hard ones as the disk responds to the driving corona (Kara and García 2025). Fourier analysis is essential for disentangling the two contributions.

Beyond this persistent stochastic variability, more extreme variability phenomena also exist, including *changing-look* AGN, which undergo dramatic changes in their broad emission lines and continuum on timescales of months to years, possibly driven by rapid changes in accretion rate or obscuration (LaMassa et al. 2015), as well as quasi-periodic eruptions (QPEs) and tidal disruption events (TDEs); these represent transient phenomena superimposed on the underlying stochastic variability and are not discussed further here.

[1]The viscous timescale, $t_{\rm visc} \sim r^2/\nu$, where ν is the kinematic viscosity, indicates the time for material to drift radially through the disk and increases with radius.

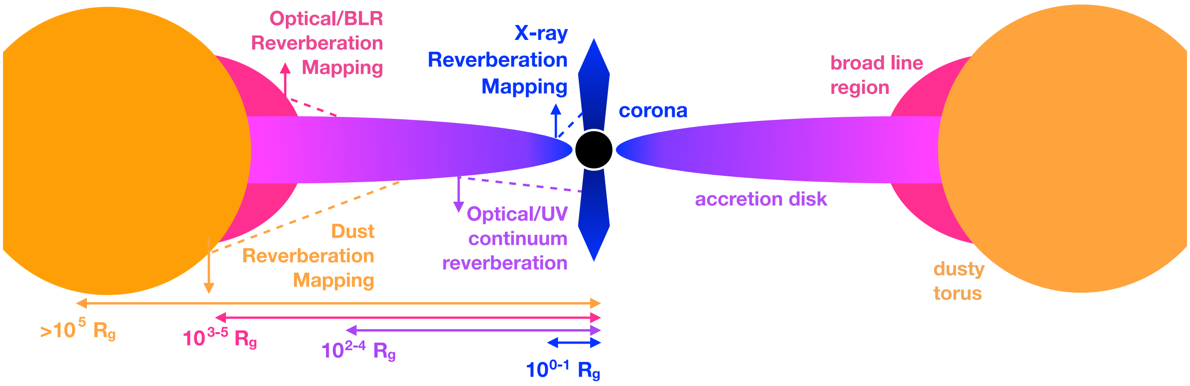


Figure 2.2: Cross-sectional schematic of an AGN depicting the four main types of reverberation mapping and their corresponding radial scales from the central black hole. Taken from Cackett et al. 2021.

2.1.1 Reverberation Mapping

The reverberation delays introduced above are not only a complication in the interpretation of light curves, they are also a powerful tool. Because the different regions of an AGN cannot be spatially resolved, their sizes cannot be measured directly, but the light-travel time between a driving source and a region that reprocesses its emission provides an indirect gauge. *Reverberation mapping* (RM) exploits this by monitoring the delay with which one part of the AGN responds to variations in another: a region at a distance R from the driving source responds on average after the light-crossing time $\tau_{\rm lag} \sim R/c$, so that measuring the lag yields the size of the region (Cackett et al. 2021). The characteristic scales and their corresponding light-crossing times for the main AGN regions are summarised in Table 2.1, ranging from hours for the X-ray corona to years for the dusty torus.

Table 2.1: Order of magnitude size scales and their equivalent light-crossing times for the different regions of an AGN. Adapted from Cackett et al. 2021.

Region	Size scale (R_g)	Light-crossing time $10^7\ M_\odot$	$10^9\ M_\odot$
X-ray corona	10	500 s	14 hr
UV/optical disk	10^2–10^4	0.06–6 days	6–600 days
BLR	10^3–10^5	0.6–60 days	60–6000 days
Dusty torus	$> 10^5$	>60 days	>6000 days

The two most established applications probe different regions. In *emission-line RM*, the broad emission lines respond to variations in the ionising continuum with a delay that measures the size of the BLR; combined with the velocity width of the lines, this yields an estimate of the black hole mass and underlies the radius-luminosity relation introduced in § 1.1.3 (Peterson 1993). In *continuum RM*, the delays are measured between different wavebands of the disk continuum itself, mapping the temperature structure of the accretion disk. For a standard thin disk the lag increases with wavelength as:

$$\tau_\lambda \propto \lambda^{4/3} \left[\frac{M_{\rm BH}\dot{M}}{(3 + f_{\rm irr})} \right]^{1/3}, \tag{2.1}$$

since longer wavelengths originate at larger, cooler radii, with the normalization set by the black hole mass, the accretion rate $\dot{M}$, and the fraction of irradiating flux $f_{\rm irr}$ (Cackett et al. 2021).

Formally, the observed response is described as the convolution of the driving light curve with a *transfer function* that encodes the geometry and kinematics of the responding region (Blandford and McKee 1982). Recovering this transfer function, or even the lag alone, is an ill-posed problem: the light curves are irregularly sampled and noisy, and different underlying signals can reproduce the same observations to within their uncertainties. This degeneracy is not unique to RM but is a general feature of inference from AGN light curves, and it motivates treating the reconstruction of a light curve probabilistically, returning a distribution of signals consistent with the data rather than a single best-fit curve. It is this probabilistic reconstruction of the individual light curve, rather than the measurement of lags between light curves, that is the subject of this thesis.

2.2 Variability Characterization

Variability has been studied under a variety of methods, starting from simple statistical quantities tracing the amplitude and timescales of the observed variability to more sophisticated approaches in both the time and frequency domains (Paolillo and Papadakis 2025). In general, these statistics attempt to measure the absolute or fractional variance of the light curve over some timescale Δt, or express the degree of correlation between different timescales or photometric bands (Ulrich et al. 1997; De Cicco et al. 2025).

The distribution of variability power over temporal frequencies (or, equivalently, 1/timescale) is quantified by the *power spectral density* (PSD), defined as the squared amplitude of the flux (e.g., Uttley et al. 2002). AGN PSDs typically follow a "red-noise" power law, $P(\nu) \propto \nu^{-\alpha}$ with $\alpha \sim 1$ at low frequencies, steepening to $\alpha \gtrsim 2$ above a characteristic break frequency ν_b, consequently displaying larger amplitude variations on longer timescales. This break timescale scales with black hole mass and accretion rate (McHardy et al. 2006), with smaller black holes varying more rapidly than bigger ones for the same observing duration (Figure 2.3). The anticorrelation can be interpreted as a consequence of the fact that the most luminous AGN are likely to host larger mass black holes (if all quasars accrete at similar Eddington rates), in which case, their variability should be diluted by the larger emitting region (Paolillo and Papadakis 2025).

The PSD also depends strongly on the observing waveband, with optical PSDs showing steeper slopes and lower normalization than X-ray PSDs, consistent with the propagating fluctuations picture where outer disk regions contribute power at lower frequencies and inner regions at higher ones (Kara and García 2025).

Both the amplitude and the timescale of the variability correlate with the physical properties of the accreting system. The *damping timescale* τ, which determines when the flux decorrelates from its past, increases with black hole mass and, more weakly, with rest-frame wavelength (e.g., Collier and Peterson 2001; Burke et al. 2021) linking the variability timescale and the thermal timescale of the accretion disk. The *variability amplitude*, in turn, decreases towards longer wavelengths, so that AGN vary with larger fractional amplitude in the bluer bands (MacLeod et al. 2010), consistent with the bluer emission arising from the hotter, more compact inner disk. These scaling relations, and in particular the dependence of τ on black hole mass, are used in Chapter 3 to assign physically motivated parameters to the simulated light curves.

A complementary characterization is provided in the time domain by the *structure function* (SF), which measures the mean variability amplitude as a function of the time lag Δt separating pairs of observations, adequate for the sparse, irregularly sampled light curves typical of optical

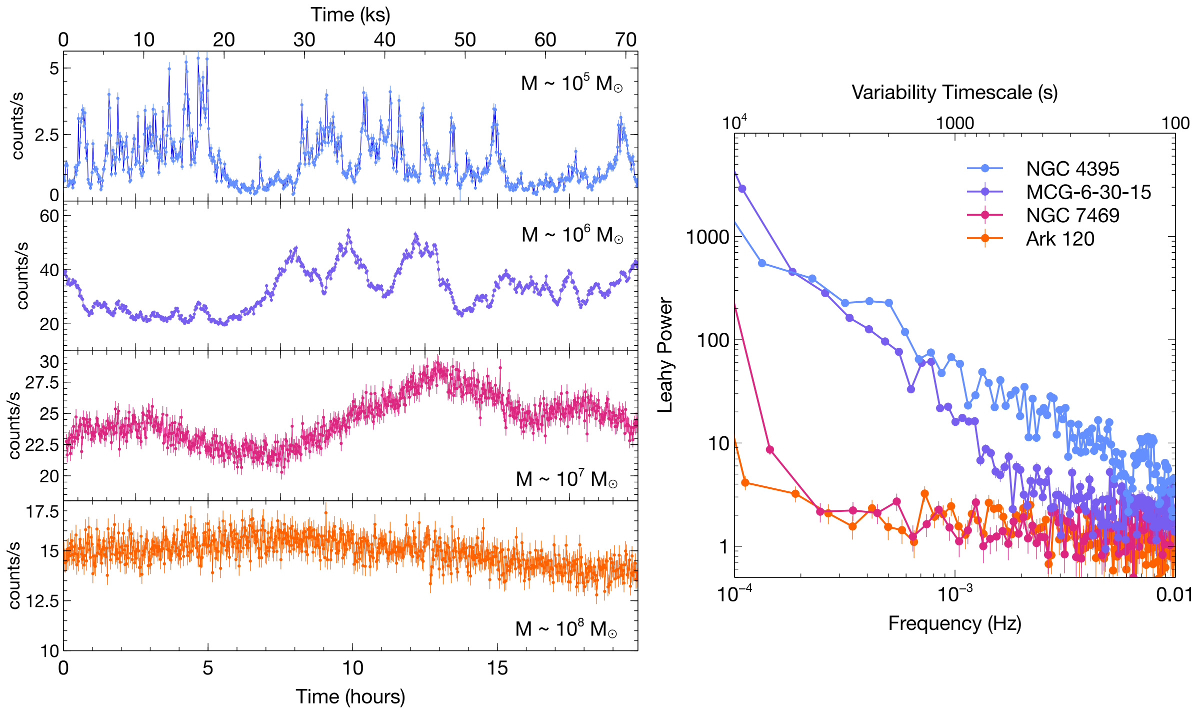


Figure 2.3: Left: X-ray light curves of four Seyfert 1 AGN with black hole masses spanning 10^5–10^8 $M_\odot$, observed with XMM-Newton over 70 ks. Smaller black holes vary more rapidly, with the 10^5 $M_\odot$ AGN (NGC 4395, top) showing variability on timescales of tens of seconds, while the 10^8 $M_\odot$ AGN (Ark 120, bottom) varies on timescales of hours. Right: The corresponding PSDs, showing the red-noise power-law shape characteristic of AGN X-ray variability. Taken from Kara and García 2025.

surveys, where a direct estimate of the PSD is difficult (Kozłowski 2016b). On short lags the SF rises as a power law, reflecting the growing amplitude of the variability, and flattens above the damping timescale τ to a constant level set by the long-term variability amplitude, mirroring in the time domain the same turnover seen as the PSD break at $\nu_b \sim 1/\tau$. The shape of the SF therefore encodes the same two main quantities that characterize the variability as a whole, an amplitude and a timescale, which are precisely the parameters of the *damped random walk* (DRW) adopted to model the light curves in the following chapter.

3 Simulation Framework

The reconstruction method developed in this thesis is trained and tested on simulated light curves, for which the underlying signal is known exactly. This chapter sets out the framework used to generate them. It begins with the stochastic process and damped random walk, derives the transition probability that the reconstruction method will later learn, and then describes the simulation itself, including the way physically motivated timescales are assigned and the sampling regime.

3.1 Stochastic Processes

The variability introduced in Chapter 2 is aperiodic and aleatory, so it is not well described by a deterministic model with a fixed set of parameters. It is instead naturally treated as a realization of a *stochastic process*: a family of random variables indexed by time, whose statistical properties, rather than whose exact values, are the object of study. Rather than predicting the flux at a given epoch, one models the distribution from which that flux is drawn, conditioned on the values already observed.

The simplest stochastic model consistent with the observed variability of AGN is the *damped random walk* (DRW), also known as an Ornstein-Uhlenbeck process (Kelly, Bechtold, and Siemiginowska 2009). It is described by the stochastic differential equation

$$df(t) = -\frac{1}{\tau} f(t)dt + \sigma_{\mathrm{d}}\sqrt{dt}\epsilon(t) \tag{3.1}$$

where τ is the damping timescale, σ_{d} sets the amplitude of the driving noise, and $\epsilon(t)$ is a white noise process[1]. The two terms admit a direct physical reading. The first is a restoring term that pulls the flux back towards its mean on the timescale τ, so that excursions do not grow without bound, while the second perturbs it with stochastic kicks. The competition between damping and driving produces the characteristic behaviour of AGN light curves, which vary coherently on timescales shorter than τ and decorrelate into noise on timescales much longer than τ.

The DRW provides a good statistical description of optical AGN light curves (Kelly, Bechtold, and Siemiginowska 2009; Kozłowski et al. 2010; MacLeod et al. 2010) and, importantly, it is a *Gaussian process*, a distribution over functions such that the values at any finite set of times follow a multivariate normal distribution, specified by a mean function and a covariance function

[1] A white noise process is one with a mean zero and no correlation between its values at different times.

(Rasmussen and Williams 2006). Conditioning on observed values yields a Gaussian predictive distribution in closed form, which is what makes such processes analytically tractable for interpolation. Its exponential autocorrelation function

$$\mathrm{ACF}(\Delta t) = \exp\left(-\frac{|\Delta t|}{\tau}\right), \tag{3.2}$$

corresponding to a power spectrum of the form $P(\nu) \propto \nu^{-2}$ at high frequencies that flattens to white noise below the break frequency $\nu_b = 1/(2\pi\tau)$.

The damped random walk is also the simplest member of the broader family of *continuous-time autoregressive moving average* (CARMA) processes (Kelly et al. 2014), corresponding to the case CARMA$(1, 0)$. A general CARMA(p, q) process is the solution of a linear stochastic differential equation of order p, driven by white noise through an order-q moving-average term, and can reproduce a wider range of power-spectrum shapes than the damped random walk alone. In particular the CARMA$(2, 1)$ process, also known as the damped harmonic oscillator, has been found to describe some observed AGN light curves more faithfully (Yu et al. 2025). The damped random walk is nonetheless retained throughout this work as the generative model, both because it remains an adequate approximation on the timescales probed by the surveys considered here and because its analytic tractability provides a controlled setting in which to develop and test the reconstruction method. The consequences of this choice, in particular the effect of model misspecification when the method is applied to real data, are returned to in Chapter 5.

3.1.1 The Transition Probability

The exponential autocorrelation of Equation 3.2 has a consequence that is central to this work. Because the damped random walk is a Gaussian process, and because Eq. 3.1 is first-order and Markovian[2], the distribution of the state at a time $t+\Delta t$ given its value at time t is itself Gaussian and depends only on the elapsed time Δt. This *transition probability* is

$$X_{t+\Delta t} \mid X_t \;\sim\; \mathcal{N}\big(\mu + e^{-\Delta t/\tau}\big(X_t - \mu\big),\; \sigma^2\big(1 - e^{-2\Delta t/\tau}\big)\big), \tag{3.3}$$

where μ is the long-term mean and σ the stationary standard deviation of the light curve. The mean relaxes exponentially from the current value towards μ on the timescale τ, while the variance grows from zero at $\Delta t = 0$ to the stationary value σ^2 as Δt becomes large compared with τ. Both limits are the expected ones. At short separations the process retains a strong memory of its previous state and the distribution is narrow and centred near X_t, whereas at separations much longer than τ the memory is lost and the distribution approaches the stationary marginal $\mathcal{N}(\mu, \sigma^2)$.

Equation 3.3 is the object that the reconstruction method of Chapter 4 learns from data. It provides both a target against which the learned transition can be validated and, through the same expression, the exact Gaussian process baseline against which the method is compared. After the light curves are centred, $\mu = 0$, the mean reduces to $X_t\, e^{-\Delta t/\tau}$ and the variance to $\sigma^2(1 - e^{-2\Delta t/\tau})$.

[2]A process is Markovian if its future depends only on its present state and not on the path by which it was reached, so that the state at one time contains all the information relevant to predicting later times.

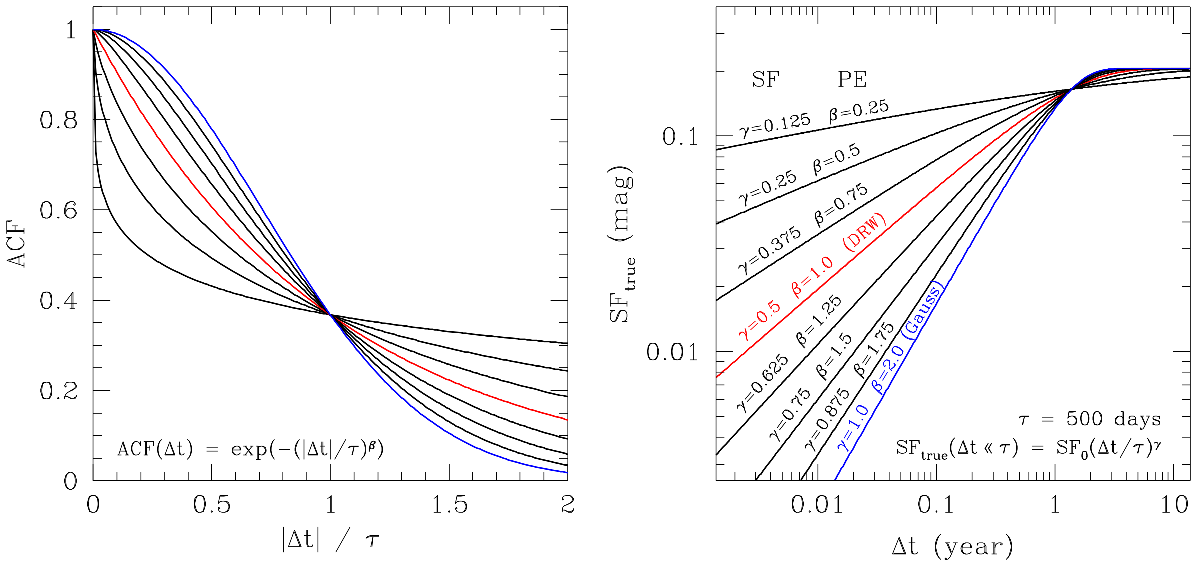


Figure 3.1: Left: autocorrelation functions of the family of Equation 3.4 for a range of the decay exponent β. Right: the corresponding structure functions, with short-timescale slope γ. The damped random walk ($\beta = 1$, red) produces a structure function of slope $\gamma = 0.5$ below the damping timescale, flattening above it, and sits between a slowly declining correlation and a Gaussian one ($\beta = 2$, blue). All curves cross at $\Delta t = \tau$, illustrating the role of the damping timescale as the characteristic decorrelation lag. Adapted from Kozłowski 2016b.

3.2 The Damping Timescale

The damping timescale is the quantity that most directly characterizes the variability, and it requires some care because its definition and its measured value depend on convention. In the structure-function representation, the damped random walk is one member of a family parametrized by the exponent of the autocorrelation decay,

$$\mathrm{ACF}(\Delta t) = \exp\left(- \left(\frac{|\Delta t|}{\tau} \right)^{\beta} \right), \tag{3.4}$$

where $\beta = 1$ recovers the damped random walk and $\beta = 2$ a Gaussian correlation. The corresponding structure function rises as a power law on short timescales, $\mathrm{SF}(\Delta t) \propto (\Delta t/\tau)^{\gamma}$ for $\Delta t \ll \tau$, before flattening above the damping timescale; the damped random walk corresponds to a short-timescale slope $\gamma = 0.5$ (Figure 3.1). Real light curves are consistent with exponents scattered around these values, which is one of the empirical justifications for adopting the damped random walk as a baseline while remaining aware of its limitations.

A second source of ambiguity is the reference frame. The damping timescale measured from an observed light curve is stretched relative to its rest-frame value by a factor of $(1 + z)$, so that comparisons between sources at different redshifts, or between measurements and physical models, must specify whether τ is quoted in the observed or the rest frame. Throughout this work τ refers to the rest-frame timescale unless stated otherwise.

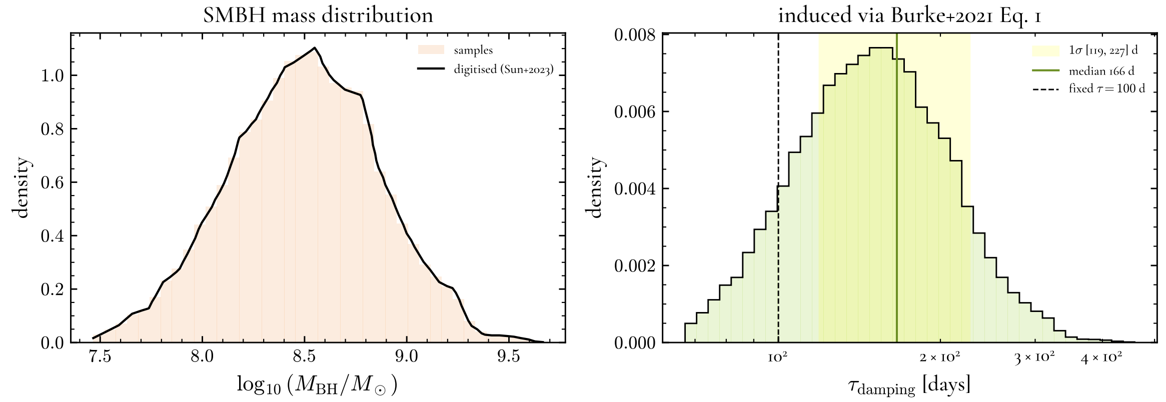


Figure 3.2: Left: the black hole mass distribution used to assign timescales, digitized from Sun 2023, with samples drawn from it by inverse-transform sampling. Right: the damping-timescale distribution induced by passing these masses through the mass-timescale relation of Equation 3.5, with median 166 d and the marked 1σ range $[119, 227]$ d. The fixed value $\tau = 100$ d used in the controlled experiment is shown for reference.

3.3 Physically Motivated Timescales

For the simulated light curves to resemble a real population, the damping timescale is not fixed at a single value but drawn from a distribution representative of observed AGN. As introduced in Chapter 2, τ increases with black hole mass, therefore timescale is assigned in two steps: first, a black hole mass is first drawn from the bias-corrected mass distribution of quasars of Sun (2023), and the corresponding damping timescale is then obtained from the mass-timescale relation of Burke et al. (2021),

$$\tau = 107^{+11}_{-12}\,\text{days}\left(\frac{M_{\rm BH}}{10^8\,M_\odot}\right)^{0.38^{+0.05}_{-0.04}}. \tag{3.5}$$

The mass distribution is digitized from the observed sample and sampled in full by inverse-transform sampling, so the resulting timescales inherit its shape through Equation 3.5, including its high-mass tail. The induced distribution has a median of $\tau \approx 166$ d, with the central 68% of curves falling between 119 and 227 d (Figure 3.2).

This procedure defines the varying-timescale configuration used for most of the experiments, in which each light curve is assigned its own τ and the reconstruction method must generalize across the range shown in Figure 3.2. A simpler configuration was also used, in which the timescale was held fixed at $\tau = 100$ d for all curves and served as a controlled setting in which the sampling and noise can be studied in isolation from any variation in τ. The varying-timescale case is the more realistic and more demanding test, and is the one against which the main results are reported.

3.4 Light Curve Generation

Light curves are generated with the EzTao package (Yu and Richards 2022), which represents the damped random walk as a Gaussian process and evaluates it through the *celerite* method (Foreman-Mackey et al. 2017). A direct evaluation of the Gaussian process likelihood requires inverting the covariance matrix at a cost that scales as $\mathcal{O}(N^3)$ in the number of epochs, which is prohibitive for

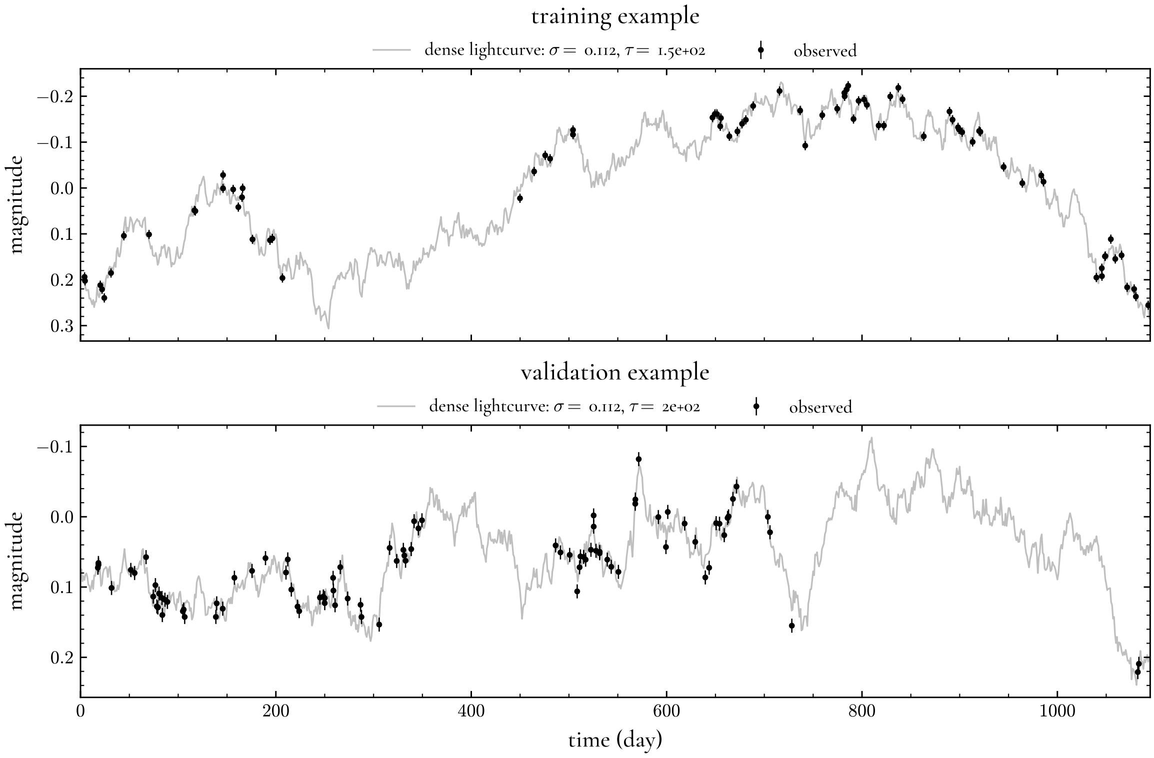


Figure 3.3: Two simulated light curves (one from the training set and one from the validation set), each showing the dense underlying realization (grey) and the sparse, noisy observations drawn from it (black).

long light curves; celerite reduces this to $\mathcal{O}(N)$ for the damped random walk kernel, making the simulation of large samples tractable. Moreover, modeling is carried out on magnitudes rather than fluxes, since the flux light curves of accreting sources are approximately lognormal and so are close to Gaussian in magnitude, matching better the assumption of the Gaussian process. For each simulated object a damping timescale is assigned as described above, the stationary amplitude is set to $\sigma = 0.112$ mag, a value representative of quasar optical variability taken from Yu et al. (2025), and a densely sampled realization is produced to represent the true, underlying signal.

The observed light curve is then formed from this dense realization rather than interpolated from it. An irregular cadence is drawn, the dense realization is evaluated on the union of the true grid and the observed epochs in a single simulation, and the observed magnitudes are read off at the sampled times and perturbed with photometric noise. This construction ensures that the observations and the underlying signal are two views of one realization, and gives access to the true signal at every epoch, which is what allows the reconstruction to be evaluated exactly. The photometric noise is added as a Gaussian perturbation at each epoch; for the simulations a single-visit uncertainty of 0.01 mag is adopted, matching the expected photometric precision of the Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory[3]. An example of each of the two configurations is shown in Figure 3.3.

[3] `https://rubinobservatory.org/for-scientists/rubin-101/key-numbers`

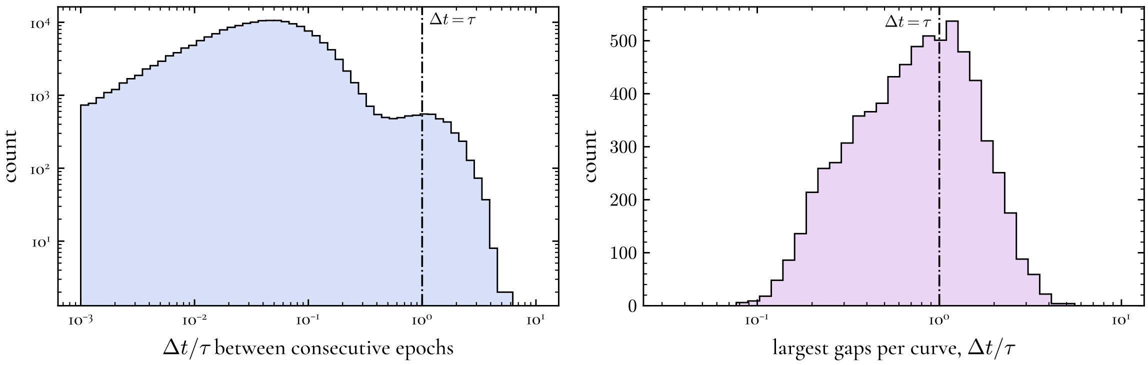


Figure 3.4: Left: the distribution of the spacing between consecutive epochs, in units of the damping timescale. Right: the distribution of the single largest gap in each light curve, in the same units. Most consecutive spacings lie well below τ, while the largest gaps straddle τ, the regime in which reconstruction is most demanding. The line marks $\Delta t = \tau$.

3.5 Sampling Regime

The difficulty of gap filling is controlled by a single dimensionless quantity, the gap length in units of the damping timescale, $\Delta t/\tau$, because a damped random walk loses memory as $e^{-\Delta t/\tau}$. The sampling parameters are therefore quoted in units of τ rather than in days. With a three-year baseline ($t_{\rm max} = 1095$ d), 130 attempted epochs, and a gap fraction of 0.35 distributed over three windows, at the fiducial timescale $\tau = 100$ d the baseline spans $10.9\,\tau$, the mean spacing between consecutive epochs is 8.4 d $= 0.084\,\tau$, and the mean gap is 128 d $= 1.28\,\tau$. Consecutive epochs are therefore strongly correlated, while the gaps straddle the decorrelation scale, the regime in which reconstruction is neither trivial nor impossible. Figure 3.4 shows the resulting distributions of the consecutive spacing and of the largest gap in each curve, both in units of τ.

This regime is bounded by two thresholds, both of which appear in the literature as biases on the recovery of τ. The timescale is underestimated when the baseline is shorter than roughly $10\,\tau$ (Kozłowski 2016a), and overestimated when it is shorter than about five times the mean cadence (Hu 2024). Here they matter for a different reason: they delimit the range over which a gap carries information at all. Below the cadence threshold consecutive epochs are effectively uncorrelated and no interpolation is possible, so a model can only return the prior; above the baseline threshold the signal barely decorrelates over the observing window and every gap is trivially interpolable. This is also why the fiducial experiments fix $\tau = 100$ d rather than adopting a broad range, since most of a wide range would produce degenerate light curves at this baseline and cadence. Where a spread of timescales is introduced, the physically motivated distribution of Section 3.3, with its bulk between roughly 100 and 230 d, lies within the informative range.

A single photometric band is used throughout. Although surveys such as ZTF and LSST observe in several bands, and although the coordinated multi-band variability discussed in Chapter 2 is in principle a rich source of additional information, the treatment here is confined to single-band light curves, and the extension to multiple bands is left as a direction for future work.

4 Neural Stochastic Flows

The reconstruction of a gappy light curve was framed in § 2.1.1 as an ill-posed problem, best approached by returning a distribution of signals consistent with the data rather than a single curve. The damped random walk introduced in Chapter 3 already provides such a probabilistic description because it is a Gaussian process and the transition between any two epochs has a known analytic form. The method adopted in this thesis replaces that transition with a flexible one learned directly from data, using *neural stochastic flows* (NSF; Kiyohara et al. 2025). The motivation is twofold, flexibility and efficiency. A learned transition can in principle capture departures from the damped random walk that a fixed kernel cannot, and, more generally, neural stochastic flows are *solver-free*: they model the transition density of a stochastic differential equation directly, without the numerical integration that conventional neural SDEs require. In the damped random walk case considered here the transition is analytically known, which provides an exact reference against which the learned model can be validated (see Chapter 5).

4.1 Normalizing Flows

A *normalizing flow* (NF) transforms a simple probability distribution, typically a standard normal $z \sim p_Z(z)$, into a more complex one through a sequence of invertible and differentiable mappings (Rezende et al. 2016; Kobyzev et al. 2021). A sample is generated by passing the base variable through the map, $y = g_\theta(z)$, the *generative direction*. Because the map is invertible, one can also go back from y to z through $z = f_\theta(y) = g_\theta^{-1}(y)$, the *normalizing direction*, and the density of y follows from the change-of-variables formula,

$$p_Y(y) = p_Z\big(f_\theta(y)\big)\,\big|\det J_{f_\theta}(y)\big|, \tag{4.1}$$

where J_{f_θ} is the Jacobian of the inverse map. The base density p_Z is known and the Jacobian term, the *volume correction*, accounts for the change of volume induced by the transformation, so Eq. 4.1 yields an exact density for y (Figure 4.1). This exactness, together with the ability both to sample and to evaluate densities, is what makes normalizing flows well suited to modelling complex distributions, and it means the model can be trained simply by maximising the likelihood of the data under Eq. 4.1 (Kobyzev et al. 2021).

For Eq. 4.1 to be practical, the transformation must satisfy two requirements: it must be easily invertible, and its Jacobian determinant must be cheap to compute (Kobyzev et al. 2021). A construction that meets both, and the one on which the model used here is built, is the *coupling flow*

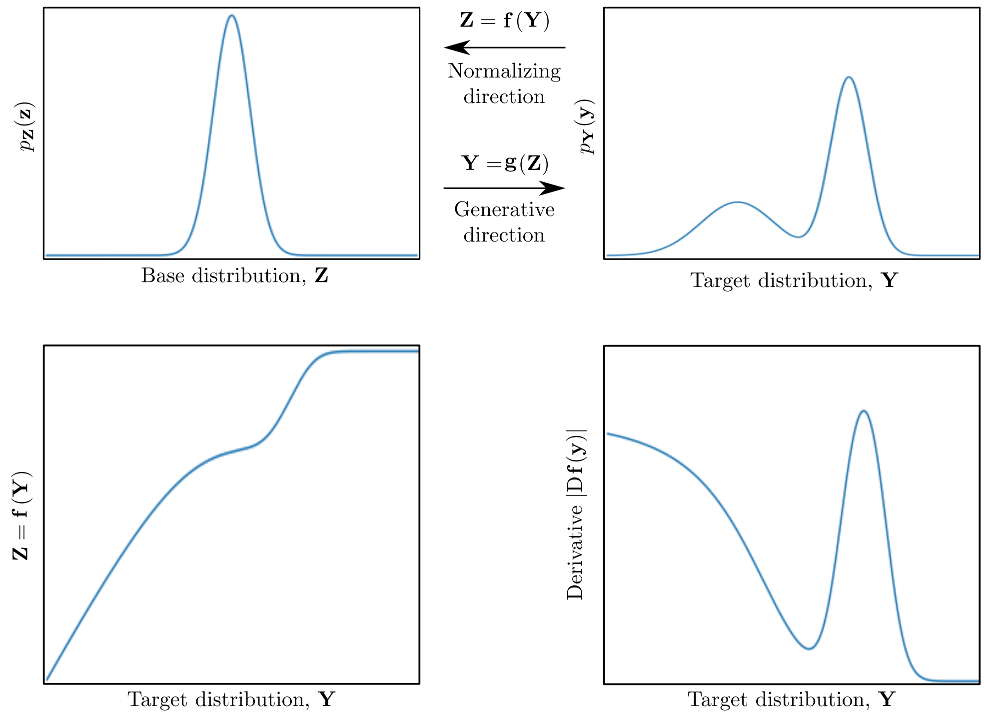


Figure 4.1: The change of variables of Eq. 4.1. Top left: the base density p_Z. Top right: the target density p_Y. A bijective map g transforms one into the other, with inverse f. Bottom left: the inverse map f. Bottom right: its absolute Jacobian, which rescales the density so that probability is conserved. Adapted from Kobyzev et al. 2021.

(Dinh, Krueger, and Bengio 2015). The input is split into two partitions; one is left unchanged, and the other is transformed by a bijection whose parameters are produced by a *conditioner* network taking the unchanged partition as input,

$$y_{\mathrm{A}} = x_{\mathrm{A}}, \qquad y_{\mathrm{B}} = x_{\mathrm{B}} \odot \exp\big(s(x_{\mathrm{A}})\big) + t(x_{\mathrm{A}}), \tag{4.2}$$

where s and t are the scale and shift functions and $\odot$ is the elementwise product. This is the affine coupling layer of RealNVP (Dinh et al. 2017). It is trivially inverted, since the unchanged partition $x_{\mathrm{A}} = y_{\mathrm{A}}$ recovers the arguments of s and t, and its Jacobian is triangular, so its determinant is simply the product of the diagonal, $\prod_j \exp(s(x_{\mathrm{A}}))_j$. Because neither the inverse nor the determinant requires inverting or differentiating s and t, these functions can be arbitrarily complex neural networks. A single such layer leaves one partition untouched, so the partitions are alternated between successive layers, and stacking many layers yields an expressive invertible map.

4.2 From Static Densities to Transition Laws

Neural stochastic flows extend this construction from a single *static* density to the *transition law* of a stochastic process (Kiyohara et al. 2025). Rather than modelling one distribution $p_Y(y)$, they model the conditional distribution of a future state given a current state and an elapsed time,

$$p_\theta\big(x_t \mid x_s, \Delta t\big), \qquad \Delta t = t - s, \tag{4.3}$$

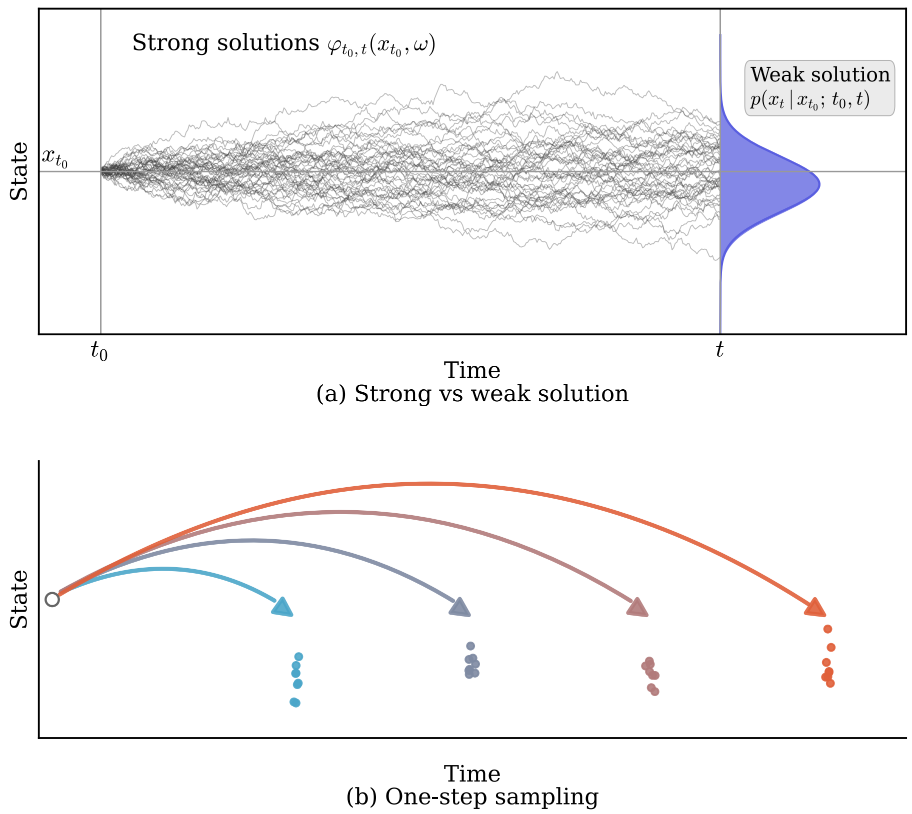


Figure 4.2: (a) The *strong solution* of a stochastic differential equation is an ensemble of paths $\varphi_{t_0,t}(x_{t_0},\omega)$ diverging from a common initial state; the *weak solution* is the transition density $p(x_t \mid x_{t_0}; t_0, t)$ they define at a later time, shown at the right edge. Neural stochastic flows model this density directly. (b) Because the density is modelled directly, samples at any elapsed time can be drawn in a single step, without the iterative integration that path-based methods require. Adapted from Kiyohara et al. 2025.

that is, the distribution of the state x_t at time t given that the process was at x_s at the earlier time s. This conditional is defined in the same way as an ordinary flow, but with the transformation now conditioned on the starting state and the elapsed time: a Gaussian noise variable $\epsilon \sim \mathcal{N}(0, I)$ is passed through a conditional normalizing flow,

$$x_t = F_\theta\big(x_s, \Delta t, \epsilon\big). \tag{4.4}$$

For fixed x_s and Δt, different draws of ϵ produce different possible future states x_t, and together they define the conditional density $p_\theta(x_t \mid x_s, \Delta t)$ of Eq. 4.3. Modelling this density directly, rather than the individual paths of the process, is what allows a state at any elapsed time to be sampled in a single step, without iterative integration (Figure 4.2).

Concretely, the sample is drawn by first initializing a state-dependent Gaussian that mirrors an Euler-Maruyama step, with a mean displaced from x_s by a drift scaled by Δt and a variance scaled by $\sqrt{\Delta t}$,

$$z = \underbrace{x_s + \Delta t\, \mathrm{MLP}_\mu(c)}_{\mu(c)} + \underbrace{\sqrt{\Delta t}\, \mathrm{MLP}_\sigma(c)}_{\sigma(c)} \odot \epsilon, \qquad c := (x_s, \Delta t), \tag{4.5}$$

and then passing it through a sequence of coupling layers of the form of Eq. A.2, whose scale and shift functions are now additionally conditioned on c and multiplied by Δt. The map F_θ

is constrained in this way so that the transition density inherits the defining properties of the solution of a stochastic differential equation (Kiyohara et al. 2025). Two are relevant here. The *identity* property requires that a zero time gap returns the input state, $p_\theta(x_t \mid x_s, 0) = \delta(x_t - x_s)$; the explicit factors of Δt in Equations A.1 and A.2 enforce this, since both the Gaussian and every coupling layer collapse to the identity as $\Delta t \to 0$. The *stationarity* property requires that, for a time-homogeneous process, the transition depend only on the gap Δt and not on absolute time; since the damped random walk is autonomous, the conditioner omits absolute time and depends on Δt alone. A third property, discussed in the following section, ties transitions of different lengths together.

This learned transition plays the role of the damped random walk step in place of the analytic transition of the Ornstein-Uhlenbeck process (Chapter 3), the flexible map F_θ models the variability between any two epochs, while reducing to the correct transition when the data are generated by a damped random walk.

4.3 Triplets and Stochastic Bridges

Reconstructing a gap requires the distribution of a state that lies *between* two observations, not merely one that follows from a single earlier state. This intermediate conditional is modelled by a *stochastic bridge*. Following the training strategy of Kiyohara et al. (2025), from each light curve one randomly samples triplets of times (t_i, t_j, t_k) with $t_i < t_j < t_k$, and two models are trained jointly on them. The *flow* model learns the transitions between states, both the direct endpoint transition $p_\theta(x_k \mid x_i, \Delta t_{ik})$ and the two one-step transitions $p_\theta(x_j \mid x_i, \Delta t_{ij})$ and $p_\theta(x_k \mid x_j, \Delta t_{jk})$. The *bridge* model learns the intermediate conditional[1]

$$q_\xi\big(x_j \mid x_i, x_k, t_i, t_j, t_k\big), \tag{4.6}$$

the distribution of the middle state given both surrounding states and their times. It is this bridge, conditioned on the two bracketing observations, that performs the gap filling at inference.

The two models are tied together by the third inherited property, the *flow property*, which requires that the transition density satisfy the Chapman-Kolmogorov relation: the direct transition from x_i to x_k must equal the composition of the two one-step transitions through any intermediate state,

$$p_\theta\big(x_k \mid x_i, \Delta t_{ik}\big) = \int p_\theta\big(x_k \mid x_j, \Delta t_{jk}\big)\, p_\theta\big(x_j \mid x_i, \Delta t_{ij}\big)\, \mathrm{d}x_j. \tag{4.7}$$

Equivalently, the joint distribution of the pair (x_j, x_k) given x_i can be factorized either as a one-step flow from x_i to x_k followed by the bridge that places x_j between them, or as a two-step flow $x_i \to x_j \to x_k$; both describe the same distribution and must agree. Enforcing this relation is what forces the flow and the bridge to represent a single, self-consistent stochastic process rather than two unrelated conditionals.

The two sides of Eq. 4.7 cannot be compared directly, since the two-step side requires marginalizing over the intermediate state x_j. The bridge q_ξ is therefore introduced as an auxiliary variational distribution, which makes the comparison tractable through variational upper bounds on the Kullback-Leibler divergence between the one-step and two-step transitions, taken in both

[1] The bridge is denoted b_ξ in the original formulation of Kiyohara et al. (2025); it is written q_ξ here to emphasize its role as an auxiliary variational distribution and for consistency with the notation used throughout this work.

directions since the divergence is asymmetric. The forward bound, from the one-step to the two-step transition, is

$$\mathcal{L}_{\text{1-to-2}} = \mathbb{E}\left[\log \frac{p_\theta(x_k \mid x_i)\, q_\xi(x_j \mid x_i, x_k)}{p_\theta(x_k \mid x_j)\, p_\theta(x_j \mid x_i)}\right], \tag{4.8}$$

with the expectation over $p_\theta(x_k \mid x_i)\, q_\xi(x_j \mid x_i, x_k)$, and the reverse bound is

$$\mathcal{L}_{\text{2-to-1}} = \mathbb{E}\left[\log \frac{p_\theta(x_j \mid x_i)\, p_\theta(x_k \mid x_j)}{q_\xi(x_j \mid x_i, x_k)\, p_\theta(x_k \mid x_i)}\right], \tag{4.9}$$

with the expectation over $p_\theta(x_j \mid x_i)\, p_\theta(x_k \mid x_j)$. Their sum is the flow-consistency loss, $\mathcal{L}_{\text{flow}} = \mathcal{L}_{\text{1-to-2}} + \mathcal{L}_{\text{2-to-1}}$.

The derivation will be explained more in depth in the Appendix A

4.4 Training Objective

The model is trained by minimising a loss with two terms (Kiyohara et al. 2025),

$$\mathcal{L}(\theta, \xi) = \mathcal{L}_{\text{data}}(\theta) + \lambda\, \mathcal{L}_{\text{flow}}(\theta, \xi), \tag{4.10}$$

where $\mathcal{L}_{\text{data}}$ is the negative log-likelihood of the observed transitions under the flow, evaluated through the change-of-variables formula of Eq. 4.1, and $\mathcal{L}_{\text{flow}}$ is the flow-consistency term of Equations A.4 and A.5, weighted by a hyperparameter λ. The data term anchors the model to the observed light curves, while the consistency term enforces that the learned transitions compose correctly across time. Both terms are necessary: the consistency term alone admits self-consistent but arbitrary transitions, and it is the data likelihood that ties the learned process to the damped random walk actually present in the light curves. The sensitivity of the results to the weighting λ is examined in Chapter 5.

4.5 Implementation

The model is implemented in JAX (Bradbury et al. 2018), building on the publicly available `jax_nsf` codebase accompanying Kiyohara et al. (2025), from which the flow and bridge architectures and the core training machinery are taken. The change-of-variables likelihood of Eq. 4.1 and the variational bounds of Equations A.4 and A.5 are differentiated automatically, so the gradients of the loss are obtained straighforward, and the full training step is compiled once and reused across iterations, which makes the repeated evaluation of the loss over many triplets efficient. The flow and bridge are represented as `equinox` modules (Kidger and Garcia 2021), and only their floating-point array leaves are treated as trainable parameters, with the remaining static structure held fixed during optimisation.

Training minimizes the objective of Eq. 4.10 using the Adam optimizer (Kingma and Ba 2017) as implemented in `optax` (DeepMind et al. 2020), with a learning rate of 2×10^{-3}. At each step a batch of sixteen light curves is drawn, and from each a set of thirty-two triplets (t_i, t_j, t_k) is sampled, with the intermediate times drawn log-uniformly so that both short and long gaps are represented. The loss is averaged over the batch and its gradient taken through the compiled step. The zero-padding introduced when the ragged light curves are stored as fixed-size arrays (Chapter

3) is excluded from the loss through a mask, so that only genuine observations contribute.

The simulation of the light curves and the training of the model are carried out in two separate environments. The simulator relies on `EzTao` (Yu and Richards 2022), whose dependencies conflict with those of `jax_nsf`, so the two cannot share an environment. The simulated light curves are written to disk and read back for training, with the saved dataset acting as the only interface between the two stages. This separation keeps the generation of the training set reproducible and independent of the model, so a dataset can be regenerated or reused without retraining.

The contributions of this thesis are in the surrounding framework rather than in the flow architecture itself: the simulation of damped random walk light curves with physically motivated parameters (Chapter 3), the construction of a training set matched to the sampling and noise of a real survey (Chapter 5), and the evaluation of the learned model against the analytic transition and a Gaussian process baseline.

5 Reconstructing AGN Light Curves

This chapter reports the performance of the reconstruction method. It proceeds from the most controlled setting to the most realistic. A fixed-timescale, low noise experiment first establishes that the method learns the analytic damped random walk transition and reconstructs light curves close to the theoretical, optimal case. A varying-timescale experiment then introduces a distribution of damping timescales. The sampling regime of the Zwicky Transient Facility (ZTF) is then measured and reproduced, and it is in this regime, with higher and epoch-dependent noise, that a limitation appears in the way the model treats measurement error. The method is finally applied to real ZTF light curves through a masking experiment.

5.1 Experimental Setup

Three configurations of the simulation framework of Chapter 3 are used, in increasing order of difficultness. In the first one, the damping timescale is held fixed at $\tau = 100$ d and the photometric uncertainty is a constant 0.01 mag against a variability amplitude of $\sigma = 0.112$ mag, giving a uniform ratio $y_{\mathrm{err}}/\sigma = 0.089$ at every epoch. In the second, the timescale is instead drawn from the physically motivated distribution of Section 3.3, while the noise is kept at the same low level. In the third, the sampling and the epoch-dependent noise are matched to those of ZTF, with the uncertainty at each epoch drawn from the survey's own error distribution.

Three quantities are used to check the reconstructions. The *coverage* is the fraction of held-out points that fall within a predictive interval of a given nominal probability [1]. The *root-mean-square error* (RMSE) measures the accuracy of the point reconstruction, taken as the posterior median, against the truth. The *predictive log-likelihood*[2] is the mean log-density assigned to the held-out points by the predictive distribution; unlike coverage it cannot be improved simply by widening the intervals, since intervals that are too wide are penalized as well as those that are too narrow. Coverage and RMSE therefore separate the two ways a reconstruction can fail, in its uncertainty and in its accuracy, while the predictive log-likelihood combines them.

Performance is measured against a Gaussian process with the damped random walk kernel. In the low noise experiments this baseline is given the true parameters of each curve, in which case it is the Bayes-optimal predictor: no method conditioned on the same observations can achieve

[1] A well-calibrated model should reproduce the nominal level, so that a central 68% interval contains the truth 68% of the time, and a coverage below the nominal level indicates overconfident, too-narrow intervals.

[2] The mean Gaussian log-density of the held-out values under the predictive distribution.

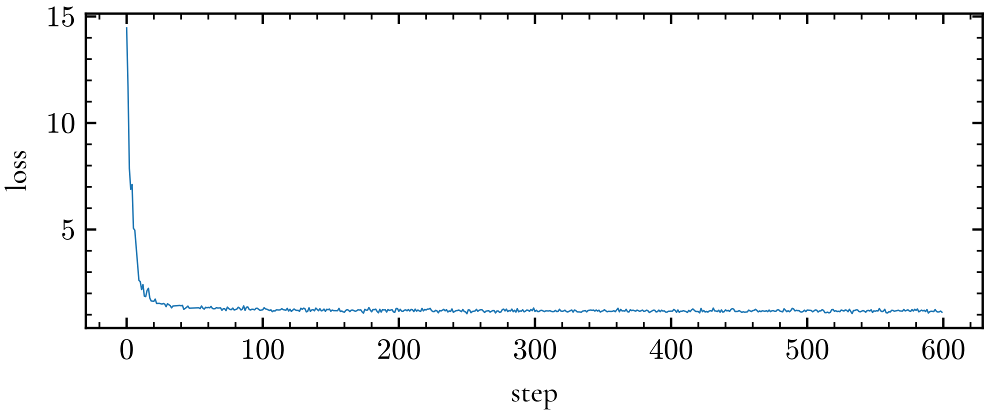
a lower expected error, so it sets the ceiling that the reconstruction is measured against. On real data, where the true parameters are unknown, the Gaussian process is instead given parameters fitted to the observations, and a linear interpolation is added as a trivial reference that provides a point prediction with no associated uncertainty.

5.2 Low Noise Experiment

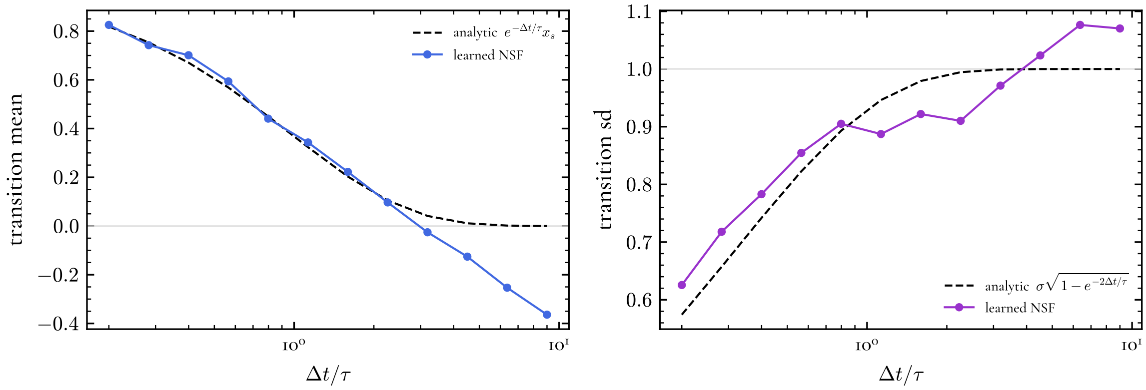
5.2.1 Fixed Timescale

The most controlled experiment fixes the damping timescale at $\tau = 100$ d for every light curve. The training objective converges within a few hundred steps (Figure 5.1a).

The first quantity examined is the learned transition itself, independently of any reconstruction. The trained flow defines a transition density $p_\theta(x_{t+\Delta t} \mid x_t, \Delta t)$, and for a range of elapsed times Δt its mean and standard deviation are obtained by sampling the flow from a fixed starting state and comparing them with the analytic damped random walk expressions of Equation 3.3.



(a) Training loss as a function of optimization step.



(b) Mean (left) and standard deviation (right) of the learned transition, in units of the damping timescale, against the analytic damped random walk (dashed).

Figure 5.1: Training and validation of the fixed-timescale model.

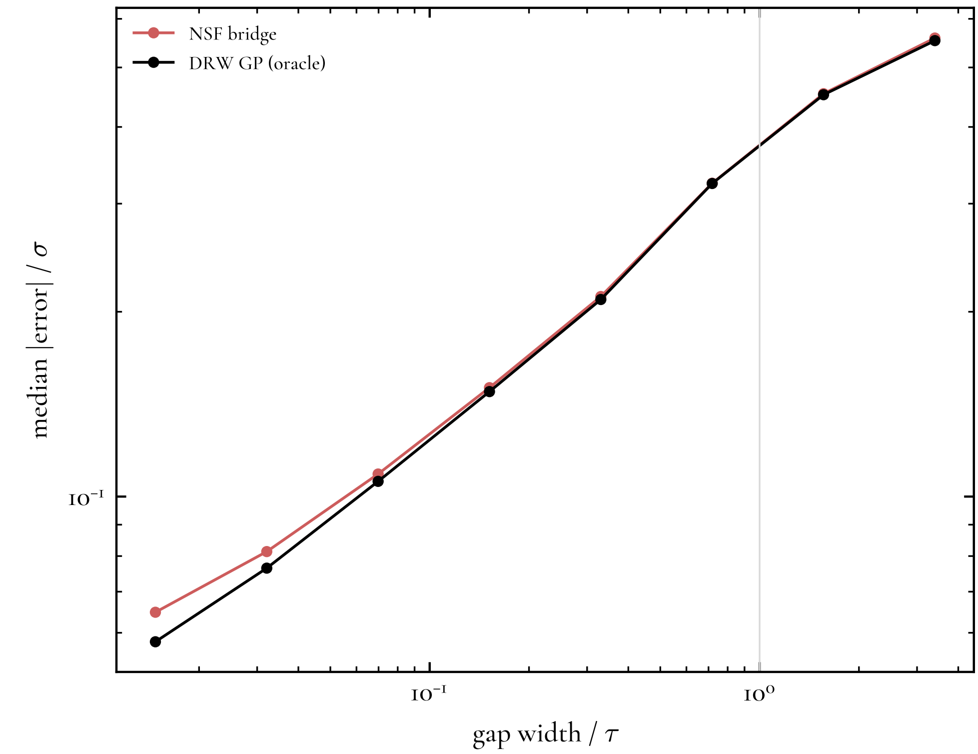


Figure 5.2: Reconstruction error as a function of gap width in units of the damping timescale, compared with the Gaussian process. The two track each other and converge at the widest gaps, where the correct prediction is the prior.

5.2.2 Varying Timescale

The timescale is next drawn from the physically motivated distribution of Section 3.3, so that each light curve carries its own damping timescale and the flow is trained on a mixture of dynamics rather than a single one. Since the transition now depends on the timescale of the individual curve, it is evaluated at the median timescale of the distribution. The learned transition remains close to the analytic one over most of the range, departing only at the longest separations, where the mean reaches -0.30 at nine damping timescales rather than zero and the standard deviation settles near 1.11 rather than unity (Figure 5.3). This residual is small and appears at the separations that are least represented in the training data, since gaps of many damping timescales are rare.

In reconstruction the method stays close to the Bayes-optimal reference. Over the varying-timescale validation set it reaches a median coverage of 0.61 at the 68% level, against 0.70 for the Gaussian process given the true parameters, and a median RMSE of 0.172 against 0.167, within roughly 3% of the optimal reference. The coverage varies with the timescale of the curve, from 0.57 on the curves with the shortest timescales to 0.64 on those with the longest. Follows from the sampling: a short timescale decorrelates faster, so for a fixed cadence the gaps span more correlation lengths and the reconstruction is harder, whereas a long timescale keeps neighbouring epochs correlated and the reconstruction is better constrained. An example reconstruction is shown in Figure 5.4.

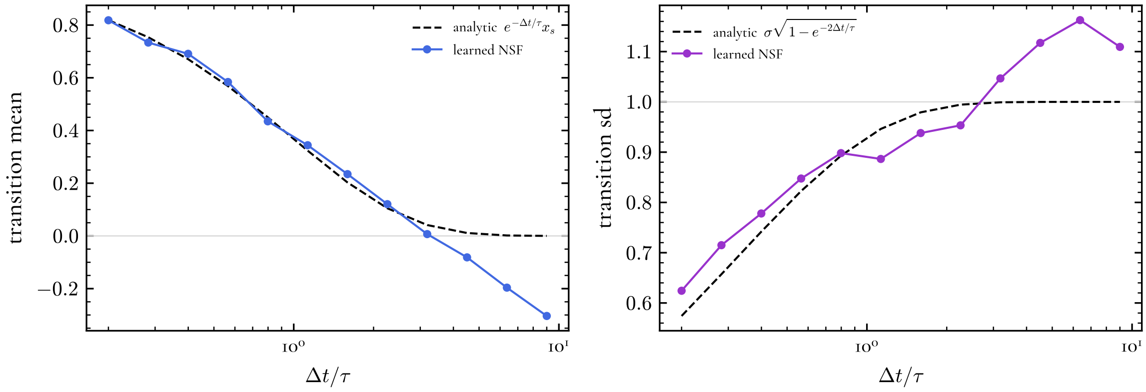


Figure 5.3: The mean (left) and standard deviation (right) of the learned transition, evaluated at the median damping timescale, compared with the analytic damped random walk (dashed). A small residual bias remains at the longest separations.

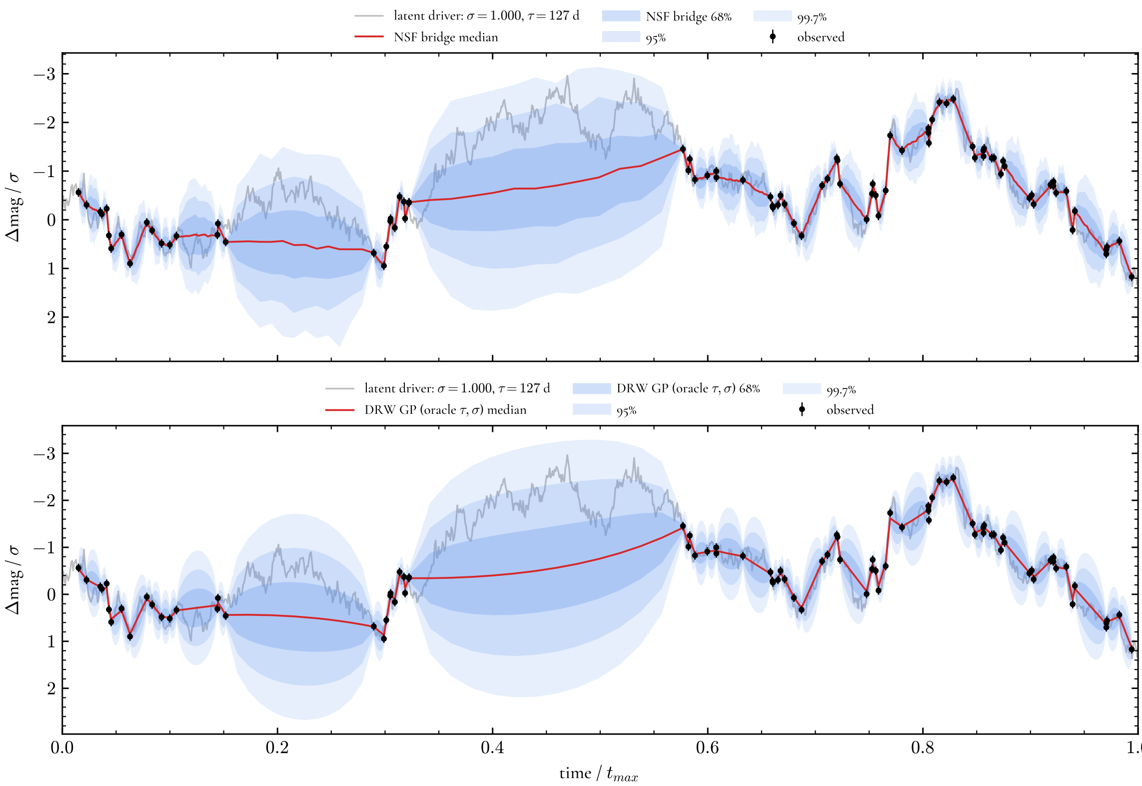


Figure 5.4: Reconstruction of a light curve from the varying-timescale validation set, with the posterior median (blue) and central 68% interval (shaded) against the true dense realization (grey).

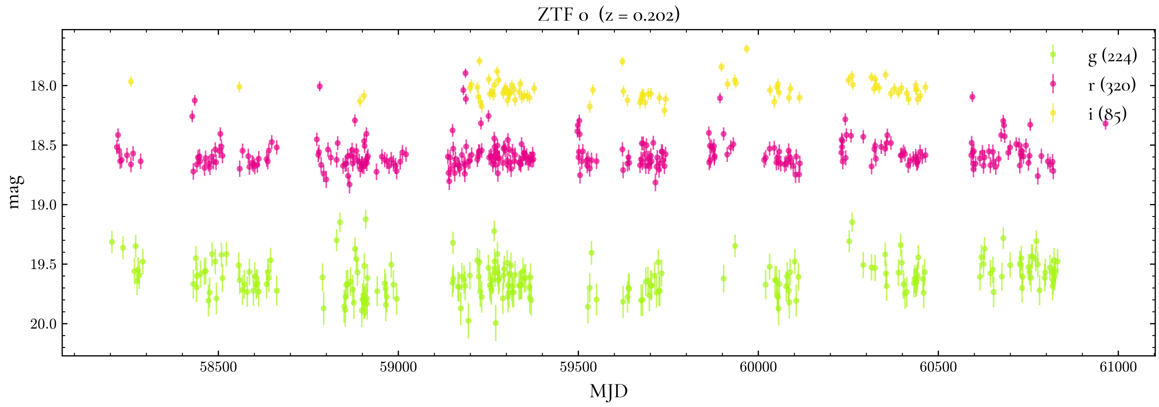


Figure 5.5: A real ZTF light curve of one AGN in the sample, showing the irregular cadence and seasonal gaps of a ground-based survey.

5.3 The ZTF Sampling Regime

The sampling regime that the simulations must reproduce is measured from a sample of real AGN. Starting from a catalogue of spectroscopically confirmed quasars, light curves are drawn from ZTF in the g, r, and i bands, and objects are retained only if they have at least 50 epochs in each band after same-night observations are combined, giving a sample of 150 AGN over a baseline of roughly seven years. An example is shown in Figure 5.5, with the irregular cadence and seasonal gaps characteristic of a ground-based survey.

The distributions of the number of epochs, the cadence, and the photometric uncertainty differ between the three bands (Figure 5.6). The r band is the most densely sampled, the g band carries the largest photometric uncertainties, since ZTF is shallower in g, and the i band is the most sparsely sampled. These per-band distributions are used directly to construct the ZTF-matched simulations, so that each simulated band inherits the cadence and noise of its real counterpart.

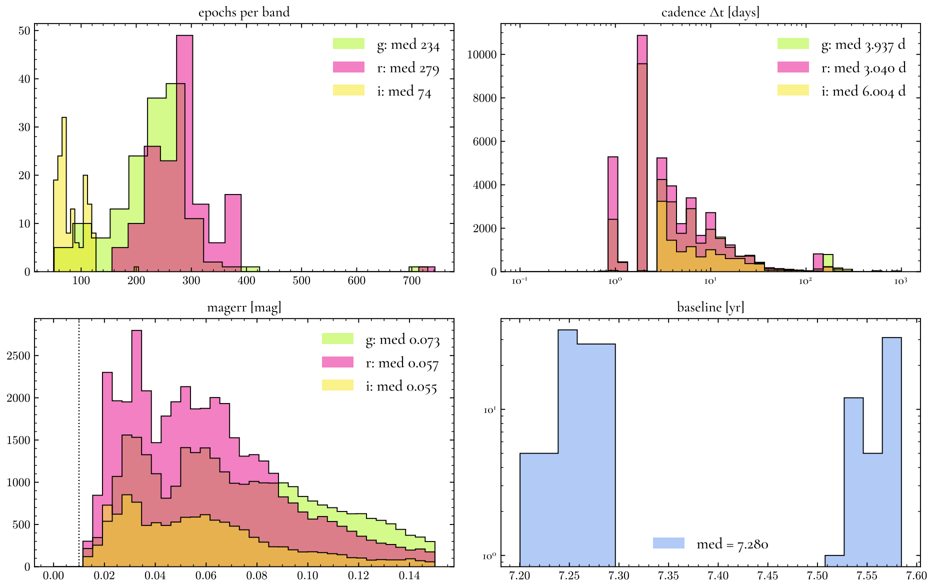
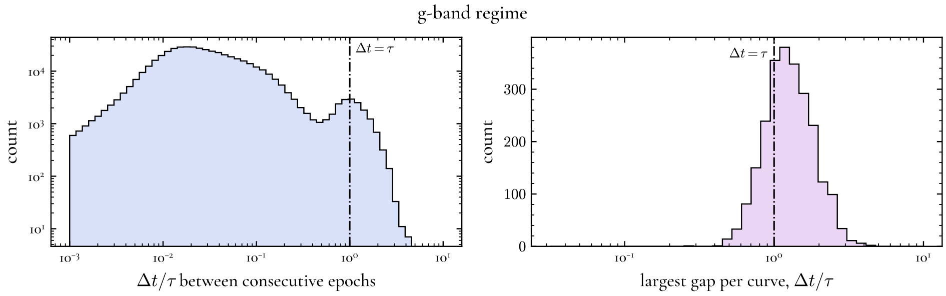
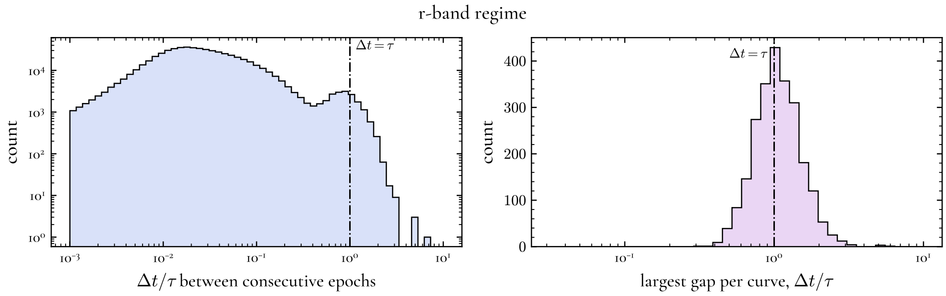


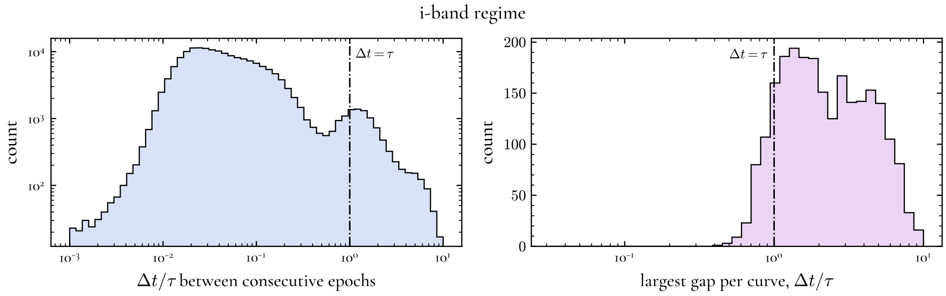
Figure 5.6: Distributions of the number of epochs, cadence, and photometric uncertainty across the 150 ZTF AGN in the g, r, and i bands.



(a) g band



(b) r band



(c) i band

Figure 5.7: The spacing between consecutive epochs (left) and the single largest gap per curve (right), in units of the damping timescale, for the ZTF-matched simulations in each band. The line marks $\Delta t = \tau$.

The quantity that controls the difficulty of reconstruction is the size of the gaps relative to the damping timescale, since a damped random walk loses memory as $e^{-\Delta t/\tau}$ and it is gaps comparable to or larger than τ that make reconstruction non-trivial. Expressed in these units the three bands probe noticeably different regimes (Figure 5.7). In the g band the median largest gap per curve is $1.24\,\tau$, with 73% of curves containing a gap wider than the damping timescale; in the r band the median largest gap is $1.05\,\tau$, with 56% exceeding τ; and in the i band the median largest gap is $2.10\,\tau$, with 89% exceeding τ and a tail reaching almost seven damping timescales. The i band therefore contains the widest gaps, some large enough that the endpoints retain essentially no information about the intervening signal, and it is the band in which single-band reconstruction is most strongly limited.

5.4 Calibration Under ZTF Noise

Trained and evaluated on light curves matched to the ZTF regime, the method continues to reconstruct the underlying signal but its predictive intervals become badly overconfident. To establish that this is a property of the model rather than of a particular training run, the training is repeated across five random seeds; the results are shown in Table 5.1. The coverage at the 68% level is far below nominal in every band, at 0.30, 0.33, and 0.42 in g, r, and i, and the scatter across seeds is small, so the overconfidence is systematic. The predictive log-likelihood worsens sharply with the noise level of the band, from -3.29 in the least noisy band to -10.63 in the noisiest, since the noisiest band produces the narrowest intervals relative to the scatter of the truth and is penalized most heavily for the points that fall outside them.

Table 5.1: Median coverage at the 68% level, RMSE, and predictive log-likelihood on the ZTF-matched validation set, as mean and standard deviation across five training seeds.

Band	Coverage @ 68%	RMSE	Predictive LL
g	0.301 ± 0.026	0.569 ± 0.004	-10.63 ± 2.34
r	0.331 ± 0.020	0.456 ± 0.003	-8.84 ± 1.43
i	0.419 ± 0.020	0.466 ± 0.002	-3.29 ± 0.44

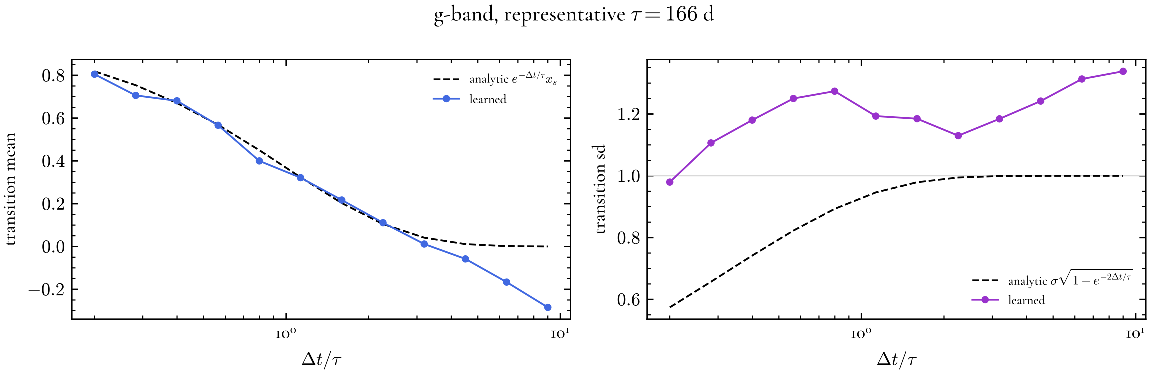
The overconfidence is present at every level of the predictive interval. Table 5.2 compares the empirical coverage of the reconstruction with that of the Gaussian process given the true parameters at four nominal levels. The Gaussian process is close to calibrated throughout, its empirical coverage matching each nominal level, while the reconstruction falls short at all of them, reaching only 0.46 where 0.90 is expected in the g band. The RMSE is a factor of roughly two larger than that of the Gaussian process in every band, so the method loses accuracy as well as miscalibrating its uncertainty.

The origin of the overconfidence is visible in the learned transition. Figure 5.8 repeats the transition comparison for models trained in the ZTF regime, in each band.

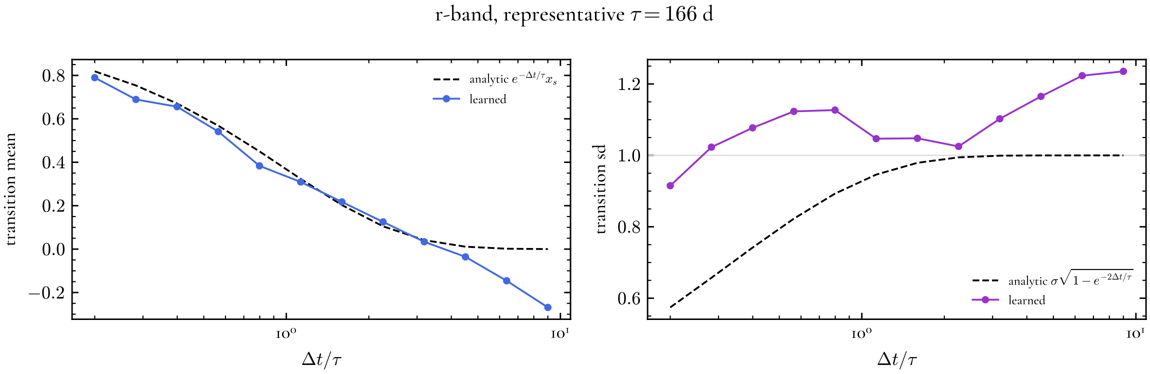
Table 5.2: Empirical coverage at four nominal levels, with RMSE, for the reconstruction and the Gaussian process given the true parameters, on the ZTF-matched validation set.

Band	Method	Coverage at nominal level				RMSE
		0.50	0.68	0.90	0.95	
g	NSF	0.21	0.31	0.46	0.51	0.572
	GP	0.51	0.68	0.89	0.94	0.296
r	NSF	0.23	0.33	0.50	0.56	0.458
	GP	0.51	0.69	0.90	0.94	0.264
i	NSF	0.31	0.43	0.61	0.67	0.470
	GP	0.50	0.67	0.89	0.94	0.318

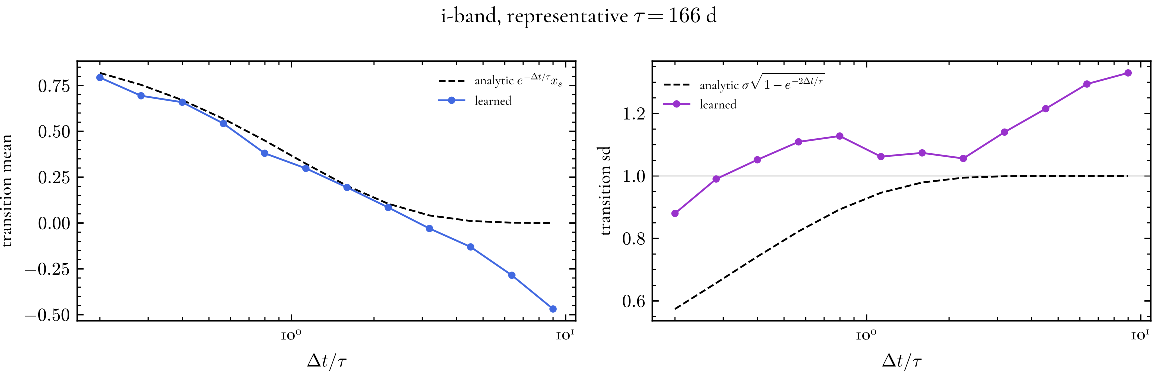
The agreement with the analytic transition is now lost, the learned mean overshoots, passing below zero instead of relaxing to it, and the learned standard deviation inflates to roughly 1.33 and fails to saturate at unity. The dynamics have not been cleanly recovered as the bridge might be anchoring too tightly to the noisy observations at the gap endpoints, treating each noisy magnitude as if it were the true signal, a reconstruction pulled towards the noise.



(a) g band



(b) r band



(c) i band

Figure 5.8: As Figure 5.1b, for models trained in the ZTF regime in each band. The learned mean overshoots past zero and the learned standard deviation inflates and does not saturate at unity.

The reconstruction error still grows smoothly with gap width (Figure 5.9)... Second, the coverage somewhat worsens as the number of epochs per curve increases (maybe add this figures). More epochs would ordinarily be expected to improve a reconstruction, but here a denser light curve supplies more noisy observations for the bridge to hold on to, not more usable information. The same effect appears across the bands, the sparsely sampled i band being the least overconfident. A model that weighted observations by their individual uncertainties would improve with

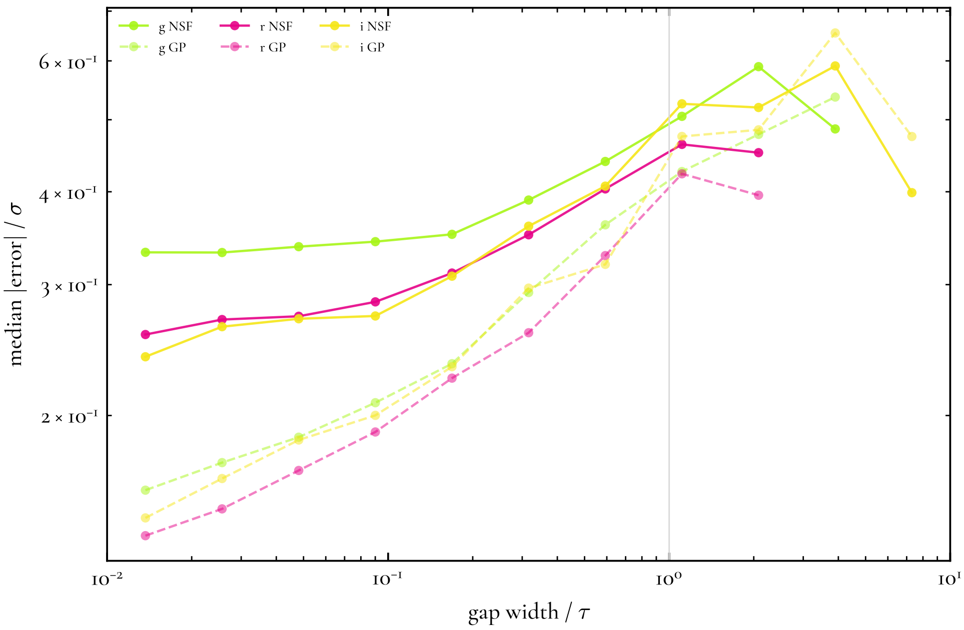


Figure 5.9: Reconstruction error as a function of gap width in units of the damping timescale, for the three ZTF bands, compared with the Gaussian process.

more epochs rather than degrade, so this dependence points directly at the absence of such a mechanism.

5.5 Application to Real ZTF Light Curves

The method is finally applied to the real ZTF light curves. Since the true signal is unavailable, the reconstruction is tested by masking. A fraction of the observations in each curve is withheld, in two forms: randomly removed isolated points, which are bracketed by nearby retained observations and test the reconstruction where it is well constrained, and points removed inside contiguous intervals, which test it across genuine gaps. Before masking, isolated outliers are removed with a local median filter, so that a single discrepant epoch is not mistaken for a real excursion. The retained points alone are used to fit the normalization and, for the Gaussian process baseline, the damping timescale and amplitude, so that the withheld observations play no part in the reconstruction. Each masked curve is reconstructed by the flow, by this fitted Gaussian process, and by linear interpolation, and the predictions are compared with the withheld observations.

Because the withheld observations are themselves noisy, a reconstruction that perfectly recovered the latent signal would still not pass through them exactly, but would scatter around them by the measurement error. The predictive intervals are therefore compared with the observations after adding the uncertainty of each withheld point to the interval in quadrature[3], so that a method calibrated against the latent signal covers the noisy observations at the nominal level.

The results are shown in Table 5.3, separately for the isolated points and for those inside gaps. On the isolated points both methods are close to calibrated, with coverage between 0.68 and 0.74,

[3]The measurement variance of a withheld point is added to the predictive variance, since the observation is a noisy realization of the latent signal the interval describes.

and the fitted Gaussian process remains close to calibrated on the harder in-gap points as well, at 0.69 to 0.75. The reconstruction instead over-covers on the in-gap points, reaching 0.85 to 0.88 against the nominal 0.68, so that across real gaps its intervals are wider than necessary. In accuracy both methods predict the withheld observations to within about one variability amplitude, the Gaussian process being consistently the more accurate; both improve on linear interpolation, and both additionally provide an uncertainty, which interpolation does not. Example reconstructions in the three bands are shown in Figure **??**.

Table 5.3: Coverage at the 68% level and RMSE for held-out prediction on real ZTF light curves, scored against the withheld noisy observations, shown separately for randomly removed isolated points and for points removed inside contiguous gaps. The Gaussian process uses parameters fitted to the retained points, and the intervals of both probabilistic methods include the uncertainty of each withheld observation. RMSE is in units of the fitted variability amplitude.

		Coverage @ 68%		RMSE	
Band	Points	NSF	GP	NSF	GP
g	isolated	0.74	0.73	0.907	0.784
	in-gap	0.88	0.74	1.019	0.963
r	isolated	0.68	0.71	0.995	0.887
	in-gap	0.85	0.75	1.080	1.012
i	isolated	0.73	0.70	0.987	0.940
	in-gap	0.85	0.69	1.101	0.999

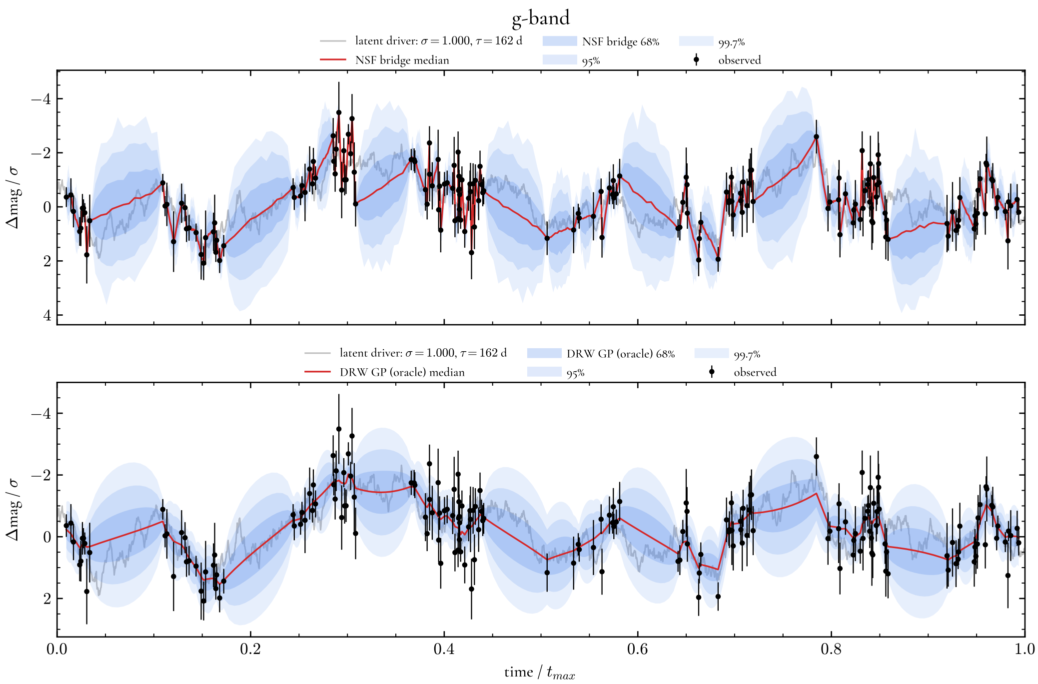


Figure 5.10: Reconstruction of a real ZTF light curve in the g band, showing the retained observations (black), the withheld observations (open), the posterior median (blue), and the central 68% interval (shaded).

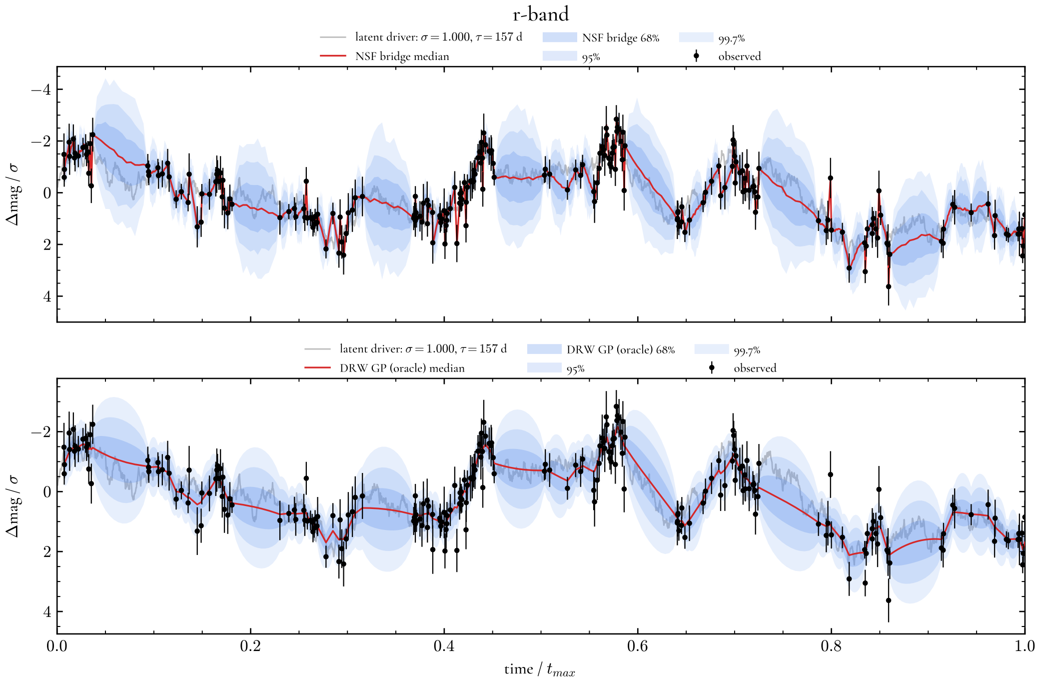


Figure 5.11: Reconstruction of a real ZTF light curve in the r band, with markers as in Figure 5.10.

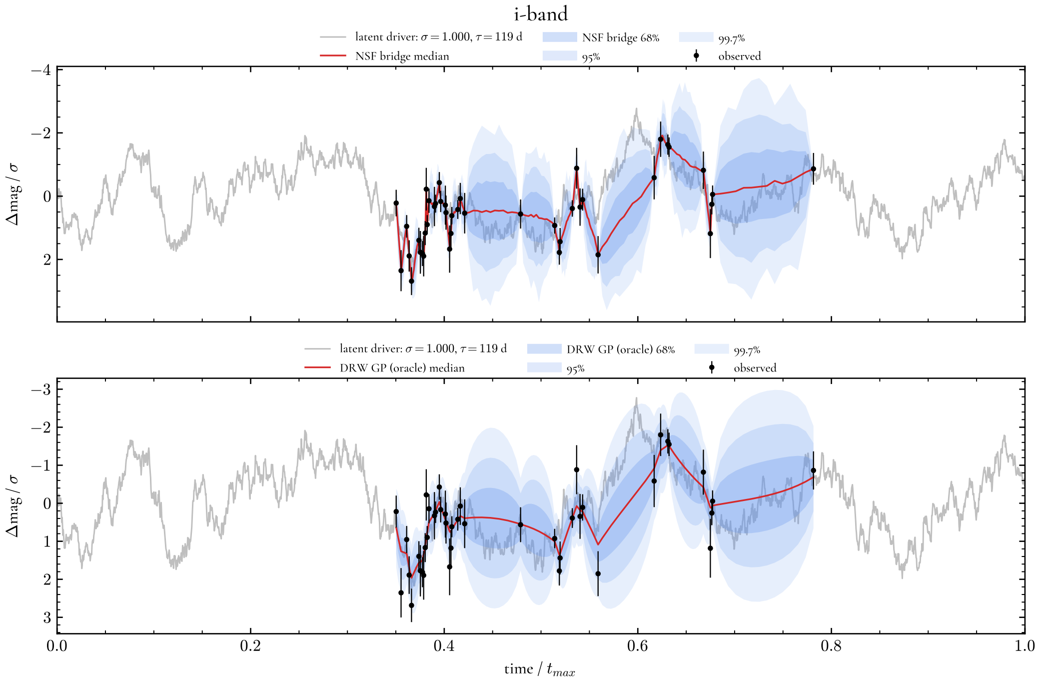


Figure 5.12: Reconstruction of a real ZTF light curve in the i band, with markers as in Figure 5.10.

The result on real data appears at first to contradict the simulated evaluation, where the reconstruction was overconfident but, in the simulated experiment, the intervals are compared with the noise-free underlying signal, while here they are compared with noisy observations and are broadened by the observational uncertainty, so the two evaluations measure the coverage of different quantities. What holds across both is that the fitted Gaussian process is a strong and well-calibrated baseline on real data, and that the calibration of the reconstruction depends on how measurement noise is treated.

5.6 Discussion

Taken together, the experiments express the limitation of the method. On low noise data, at both fixed and varying timescale, the reconstruction approaches the analytic transition and performs within a few per cent of the Bayes-optimal Gaussian process, so the model and its training objective are sound across a distribution of dynamics. The difficulty appears only when the data carry a significant noise: the reconstruction error tracks the Gaussian process and grows smoothly with gap width, while the coverage departs from nominal. The dependence of this collapse on the number of epochs, deepening a little as more noisy observations are added, together with the loss of the learned transition under noise, points to the model treating the observed magnitudes as if they were the true signal.

The discussed, natural next step is model that consumes the per-epoch uncertainties, treating each observation as a noisy measurement of a latent state rather than as the state itself and propagate that uncertainty into its predictions. Such a latent formulation, in which the flow describes the evolution of an unobserved signal with a term that relates it to the noisy observations, is the principal direction left for future work. A second direction, motivated by the widest gaps found in the i band, is the extension to multiple bands, since the coordinated variability across bands discussed in Chapter 2 would allow a gap in one band to be constrained by the others.

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A Neural Stochastic Flow in Depth

Architecture

Let $c := (x_{t_i}, \Delta t, t_i)$ denote the conditioning variables, with t_i retained only for non-autonomous systems and omitted for the autonomous damped random walk considered here, and let $\Delta t := t_j - t_i$. A sample of the transition density is drawn by first initialising a state-dependent Gaussian that mirrors the Euler-Maruyama discretisation, with drift scaled by Δt and diffusion by $\sqrt{\Delta t}$,

$$z = \underbrace{x_{t_i} + \Delta t \, \mathrm{MLP}_{\mu}(c; \theta_{\mu})}_{\mu(c)} + \underbrace{\sqrt{\Delta t} \, \mathrm{MLP}_{\sigma}(c; \theta_{\sigma})}_{\sigma(c)} \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{A.1}$$

and then passing it through a sequence of L conditioned bijective layers,

$$x_{t_j} = f_{\theta}(z, c) = f_L(\,\cdot\,; c, \theta_L) \circ \cdots \circ f_1(z; c, \theta_1). \tag{A.2}$$

Each layer splits its input into two partitions $(z_{\mathrm{A}}, z_{\mathrm{B}})$ and applies an affine update to one conditioned on the other and on c,

$$f_i(z; c, \theta_i) = \mathrm{Concat}\Big(z_{\mathrm{A}}, \; z_{\mathrm{B}} \odot \exp\big(\Delta t \, \mathrm{MLP}^{(i)}_{\mathrm{scale}}(z_{\mathrm{A}}, c)\big) + \Delta t \, \mathrm{MLP}^{(i)}_{\mathrm{shift}}(z_{\mathrm{A}}, c)\Big), \tag{A.3}$$

with the partitions alternating between layers. The explicit factor of Δt in Equations A.1 and A.3 ensures that the whole map reduces to the identity as $\Delta t \to 0$, enforcing the identity property, while the coupling structure keeps the Jacobian log-determinant tractable for the likelihood in Equation 4.1.

Flow-consistency loss

The flow-consistency term enforces the Chapman-Kolmogorov relation of Equation 4.7 by minimising the Kullback-Leibler divergence between the one-step and two-step transitions. Because this divergence is intractable directly, the bridge $q_{\xi}(x_{t_j} \mid x_{t_i}, x_{t_k})$ is introduced as an auxiliary variational distribution, and tractable variational upper bounds are minimised in both directions

(Kiyohara et al. 2025). The forward bound, from the one-step to the two-step transition, is

$$\mathcal{L}_{\text{flow,1-to-2}} = \mathbb{E}\left[\log \frac{p_\theta(x_{t_k} \mid x_{t_i})\, q_\xi(x_{t_j} \mid x_{t_i}, x_{t_k})}{p_\theta(x_{t_k} \mid x_{t_j})\, p_\theta(x_{t_j} \mid x_{t_i})}\right], \tag{A.4}$$

with the expectation taken over $p_\theta(x_{t_k} \mid x_{t_i})\, q_\xi(x_{t_j} \mid x_{t_i}, x_{t_k})$, and the reverse bound is

$$\mathcal{L}_{\text{flow,2-to-1}} = \mathbb{E}\left[\log \frac{p_\theta(x_{t_j} \mid x_{t_i})\, p_\theta(x_{t_k} \mid x_{t_j})}{q_\xi(x_{t_j} \mid x_{t_i}, x_{t_k})\, p_\theta(x_{t_k} \mid x_{t_i})}\right], \tag{A.5}$$

with the expectation over $p_\theta(x_{t_j} \mid x_{t_i})\, p_\theta(x_{t_k} \mid x_{t_j})$. The total flow-consistency loss is the sum of the two,

$$\mathcal{L}_{\text{flow}} = \mathcal{L}_{\text{flow,1-to-2}} + \mathcal{L}_{\text{flow,2-to-1}}, \tag{A.6}$$

and enters the training objective of Equation 4.10 weighted by λ.